UFR de mathématique et d'informatique
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Internship report

# The Riemann-Hilbert correspondence, $\mathcal{D}$-modules, and a bit of arithmetic periods

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I would like to thank Professor Huber for supervising this internship. Despite her busy schedule, she provided me with invaluable advice. The topic she assigned allowed me to explore the vast world of geometry—spanning analytic, algebraic, and arithmetic aspects. The warm welcome I received from her and the rest of the staff—both academic and administrative—enabled me to work under excellent conditions. In particular, I would like to thank Ben Snodgrass, Riccardo Tosi, and Ludmila Frei, whom I could turn to whenever I had questions.

*« No, no. Don't say where we are!*
*Once we know where we are, then the world becomes as narrow as a map.*
*When we don't know, the world feels unlimited. »*
Liu Cixin, *The Dark Forest*

# Contents

# I Introduction

The main goal of the mathematical content of the internship was to understand the Riemann-Hilbert correspondence at multiple levels, with the ultimate objective to relate it to number theory.
A detailed account of the internship's course can be found in II.

The *Riemann-Hilbert correspondence* takes its roots in the theories of holomorphic functions and differential equations. One of the main problems of complex functions solutions of differential equations is the multi-valuation (for instance, the complex log, which is "the solution" of $zy' = 1$ on $\mathbb{C}^*$). The theory of *monodromy* was developed to get a better grip on the behavior of such solutions. An inverse problem was the primary concern of Hilbert (along with other mathematicians of his time), described in IV.3.
The modern formulation makes use of sheaf theory (as multi-valued solutions behave well only locally, see III.2), connections (for a proper description of differential equations, see III.3.3) and category theory (ensuring the formulation itself is *canonical*).

Fuchs developed the theory in dimension one (see IV.3.1), followed by Deligne who extended it first to higher dimension (see IV.3.2), and subsequently in an algebraic setup on an algebraic variety (see V.2).
In parallel, algebraic analysis, and particularly Kashiwara, developed the theory of $\mathcal{D}$-modules to translate differential equations in terms of modules over the ring of differential operators, in a way similar to how linear algebra is used to solve linear systems. This theory works well only in the derived category setting, requiring homological algebra.
The Riemann-Hilbert correspondence finds its way into this theory, and its most refined formulation makes use of perverse sheaves.

The final part of the report is less directly related to this initial study. It deals with number theory in its fundamentals: the study of some particular complex numbers, attached to algebraic varieties. The formalism developed in the previous sections allows for a more precise study of their variations over a family of algebraic varieties, via the *Picard-Fuchs equations*.

# II Course of the internship

This report is the result of an internship I conducted at the Mathematical Institute at Freiburg im Breisgau, Germany, under the supervision of Prof. Dr. Annette Huber-Klawitter between May 25th and July 25th of 2026.
I wrote out a "blog" at the address `https://grenadine-las.github.io/site-web/feuille_de_route_memoire_fribourg.html` during the internship. I will summarize its content in this section: what I have studied, in what order, the talks I gave and the ones I attended, etc.

Before arriving in Freiburg im Breisgau, I attended to some sessions of the seminar on algebraic $\mathcal{D}$-modules co-organized by Annette Huber-Klawitter and Ben Snodgrass, see `https://home.mathematik.uni-freiburg.de/arithgeom/lehre/ss26/programm.pdf` for the program. Moreover, during a video interview, I had been given as a homework to get some knowledge on sheaf theory in order to prepare a talk in the aforementioned seminar (talk 8). It is also essential for what I was about to study: the Riemann-Hilbert ($RH$ for now on) correspondence.

The first day I arrived, after a tour of the institute and of the (few, it was a holiday week) staff, the big picture of the mathematical content of the internship had been dropped: the Riemann-Hilbert correspondence, first on a complex manifold (IV.2), using the notion of connections and local system, and then in a more algebraic way with $\mathcal{D}$-modules (VI), using perverse sheaves and derived categories. The final goal would be to relate this setup with periods coming from number theory. This is the last part of this report, see VII.3. This link is made in particular with the Gauss-Manin connection. Finally, I was given to study the specific computation for the Riemann surface $\hat{\mathbb{C}} \setminus \{s_0, \dots, s_k\}$.

The first three week were dedicated to the (re)familiarization of the terms: first some historical background to the RH correspondence, analytic and algebraic complex geometry, category theory, sheaf theory, etc. a lot of this can be found in III.
In parallel, a friend of mine, Constance, recommended me the fantastic book of T. Szamuely [Sza12], in which the RH correpondence is addressed in dimension 1, and references given for further development. I thank her for that.

I gave the talk "Background in sheaf theory" and was asked to do the 10th one entitled "Riemann-Hilbert correspondence": this was the one I was given as a particular example. I had some trouble understanding the meromorphic lifting of connections, as expressed in the fundamental work of Deligne [Del70]. See [Kat76] for an introduction.

After this, it was time to switch in an algebraic setup. First, study the correpondence in terms of varieties over $\mathbb{C}$, using the GAGA machinery.
Then in the language of $\mathcal{D}$-modules, stated in the derived categories. The whole theory, with the one related to (co)homology, took me some time to process. Everything needed to be translated in these terms, and for example, the RH correspondence as stated in [HTT08] only appears Chapter 7, knowing that some contents in every previous chapter was necessary. I still did not make it through every detail ; that is one reason why this section is so sketchy in the report.

Finally, the arithmetic-geometry-related part with periods. The first thing to understand was the Gauss-Manin connection, its different constructions and how it was important for the study of periods around singularities. I only understood few of these results: I had to set aside the theoretical aspect, as I have not found anyone who has written down the details (except for particular case as the Legendre family of elliptic curves). The horizon in this direction seems wide, and some of it will be discussed in the conclusion.

Regarding the content of the report, it aimed at being self-contained, and giving references the most precise possible. It might be longer than expected, but I tried to put everything that I saw during the internship, including the prerequisites, that I had to go through to understand the rest of the content.

Aside, I had the chance to participate to some conferences: three sessions of the "algebra seminar" at the Institute, the "BNS" (for *basic notion seminar*), a colloquium, and back in Strasbourg, the ninth meeting of the "Rhine Seminar on Transcendence", and finally the 60th year anniversary of the IRMA.

# III Prerequisites

## III.1 Category theory

For the first three subsubsection, we follow the presentation by Assem in [Ass22] and try to summarize it for the purpose of this paper. We add the precise reference for the aforementioned book to each paragraph and each result that would need a proof.
Of course, we expose the basic notions of categories, so many books contain such material.

### III.1.1 First notions

#### III.1.1.a. Starting point

[[Ass22], Chatper I].

**Definition III.1.**

A *category* $\mathscr{C}$ is given by the data of:

1. a class $\mathrm{Ob}(\mathscr{C})$, called the *class of objects* of $\mathscr{C}$,
2. for every couple of objects $(x, y)$ of $\mathscr{C}$, a class $\mathrm{Hom}_{\mathscr{C}}(x, y)$ , called the *class of morphisms* from $x$ to $y$,
3. for every triplet of objects $(x, y, z)$ of $\mathscr{C}$, a function

$$\mathrm{comp}_{(x,y,z)}: \begin{array}{ll} \mathrm{Hom}_{\mathscr{C}}(x,y) \times \mathrm{Hom}_{\mathscr{C}}(y,z) & \longrightarrow \mathrm{Hom}_{\mathscr{C}}(x,z) \\ (f,g) & \longmapsto g \circ f \end{array}$$

   satisfying the following proprerties:

   a. (*Associative law*) for every quartet of objects $(w, x, y, z)$ of $\mathscr{C}$, if $(f, g, h) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z)$, then $(h \circ g) \circ f = h \circ (g \circ f)$,

   b. (*Identities*) for every object $x$ of $\mathscr{C}$, there exists an element $1_x$ of $\mathrm{Hom}_{\mathscr{C}}(x, x)$, so that for every couple of objects $(w, y)$ and every morphisms $(f, g) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y)$, then $1_x \circ f = f$ and $g \circ 1_x = g$.

   $g \circ f$ is called the *composition* of $g$ with $f$.

**Remark III.1.**

- We will use the following notations: $x \in \mathrm{Ob}(\mathscr{C})$ if $x$ in an object of $\mathscr{C}$, $f : x \to y$ if $f$ is an element of $\mathrm{Hom}_{\mathscr{C}}(x, y)$.
- If $f : x \to y$, $x$ is called the *domain* or *source* of $f$ and $y$ the *codomain* or *target* of $f$.
- For our purpose, it is sufficient to consider that for every $(x, y) \in \mathrm{Ob}(\mathscr{C})^2$, $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is a set. In this case, the category is said to be *locally small*.

**Example III.2.**

We only give examples that will be useful in this text. For an extensive list of examples, see [[Ass22] I. Examples 2.2].

1) $\mathscr{S}et$, $\mathscr{A}b$, $\mathscr{R}ings$ will respectively denote the category of sets, abelian groups, commutative

rings with unit, with the usual associated morphisms.

2) Let $(X, \mathcal{T})$ be a topological space. We define its topology category $\mathcal{T}_X$ having objects $\mathrm{Ob}(\mathcal{T}_X) = \mathcal{T}$ and morphism $\mathrm{Hom}_{\mathcal{T}_X}(U,V) = \begin{cases} U \hookrightarrow V & \text{if } U \subseteq V \\ \emptyset & \text{otherwise.} \end{cases}$

**Definition III.2.**

Let $\mathscr{C}$ be a category.
A category $\mathscr{D}$ is said to be a *subcategory* of $\mathscr{C}$ if the following are satisfied:

1. every object of $\mathscr{D}$ is an object of $\mathscr{C}$,
2. for every $(x,y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x,y) \subseteq \mathrm{Hom}_{\mathscr{C}}(x,y)$,
3. for every $x \in \mathrm{Ob}(\mathscr{D})$, then $1_x \in \mathrm{Hom}_{\mathscr{D}}(x,x)$,
4. for every $(x,y,z) \in \mathrm{Ob}(\mathscr{D})^3$, every $(f,g) \in \mathrm{Hom}_{\mathscr{D}}(x,y) \times \mathrm{Hom}_{\mathscr{D}}(y,z)$, then $g \circ f \in \mathrm{Hom}_{\mathscr{D}}(x,z)$.

$\mathscr{D}$ is said to be a *full subcategory* of $\mathscr{C}$ if in addition, for every $(x,y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x,y) = \mathrm{Hom}_{\mathscr{C}}(x,y)$.

### III.1.1.b. Functors

A functor is to be thought as a "function between categories".

**Definition III.3.**

Let $\mathscr{C}$ and $\mathscr{D}$ be two categories.
A functor[a] $F : \mathscr{C} \to \mathscr{D}$ is the data of

1. for every $x \in \mathrm{Ob}(\mathscr{C})$, an object $F(x) \in textOb(\mathscr{D})$,
2. for every morphism $f : x \to y$ in $\mathscr{C}$, a morphism $F(f) : F(x) \to F(y)$ verifying:
   a. for every $x \in \mathrm{Ob}(\mathscr{C}), F(1_x) = 1_{F(x)}$,
   b. if $x \xrightarrow{f} x \xrightarrow{g} z$ are two morphisms in $\mathscr{C}$, then $F(gf) = F(g)F(f)$.

[a]sometimes called covariant functor

**Example III.3.**

If $\mathscr{D}$ is a subcategory of $\mathscr{C}$, the *forgetful functor* $\mathrm{Forget} : \mathscr{D} \to \mathscr{C}$ is the inclusion functor.

### III.1.1.c. Natural transformations

Natural transformation are "functions between functors":

**Definition III.4.**

Let $F, G : \mathscr{C} \to \mathscr{D}$ two functors.
A *natural transformation* $\varphi : F \to G$ is the data of $\varphi_x : F(x) \to G(x)$ in $\mathscr{D}$ for each $x \in \mathrm{Ob}(\mathscr{C})$, such that for every $f : x \to y$ in $\mathscr{C}$, we have

$$G(f) \circ \varphi_x = \varphi_y \circ F(f).$$

**Definition III.5.**

A natural transformation $\varphi : F \to G$ is said to be a *natural isomorphism* if there exists a natural transformation $\psi : G \to F$ such that $\psi \circ \varphi = 1_F$ and $\varphi \circ \psi = 1_G$, for clear composition and identity of functors.
If there exists a natural isomorphism between $F$ and $G$, we denote $F \cong G$.

Finally, we define the central notion of equivalent categories:

**Definition III.6.**

Two categories $\mathscr{C}$ and $\mathscr{D}$ are said to be *equivalent* if there exists $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$ such that $FG \cong 1_{\mathscr{D}}$ and $GF \cong 1_{\mathscr{C}}$.

### III.1.2 Additive categories

**Definition III.7.**

A category $\mathscr{C}$ is said to be a *additive* if:

1. for every $x, y \in \mathrm{Ob}(\mathscr{C})$, the set $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is equipped with the structure of an abelian group,
2. the composition is left and right distributive over the sum,
3. it admits finite products and coproducts.

The good functors in additive categories are the following:

**Definition III.8.**

Let $F : \mathscr{C} \to \mathscr{D}$ a functor between additive categories.
$F$ is said to be *linear* if for every $f, g \in \mathrm{Hom}_{\mathscr{C}}(X, Y)$, we have

$$F(f + g) = F(f) + F(g).$$

### III.1.3 (Co-)limits

The theory of (co)limits is way more wide that what we present here. We again restrict to our purpose.

#### III.1.3.a. (Co-)kernels

**Definition III.9.**

Let $\mathscr{C}$ be an additive category, $f : X \to Y$ a morphism.

1) A *kernel* of $f$ is a couple $(U, u)$ with $U \in \mathrm{Ob}(\mathscr{C})$ and $u : U \to X$ such that:
   a. $fu = 0$,
   b. for every couple $(U', u')$ with $U' \in \mathrm{Ob}(\mathscr{C})$ and $u' : U' \to X$ satisfying $fu' = 0$, there exists a unique $g : U' \to U$ so that $ug = u'$:

$$\begin{tikzcd} U \arrow[r, "u"] & X \arrow[r, "f"] & Y. \\ U' \arrow[u, dashed, "\exists ! g"] \arrow[ur, "u'"'] & & \end{tikzcd}$$

Such a couple is often denoted by $\mathrm{Ker}(f)$.

2) A *cokernel* of $f$ is a couple $(V, v)$ with $V \in \mathrm{Ob}(\mathscr{C})$ and $v : Y \to V$ such that:
   a. $vf = 0$,
   b. for every couple $(V', v')$ with $V' \in \mathrm{Ob}(\mathscr{C})$ and $v' : V' \to X$ satisfying $v'f = 0$, there exists a unique $g : V \to V'$ so that $gv = v'$:

$$\begin{tikzcd} X \arrow[r, "f"] & Y \arrow[r, "v"] \arrow[dr, "v'"'] & V \arrow[d, dashed, "\exists! g"] \\ & & V'. \end{tikzcd}$$

Such a couple is often denoted by $\mathrm{Coker}(f)$.

#### III.1.3.b. Canonical factorisation

See [Ass22, VII §4].

**Definition III.10.**

Let $\mathscr{C}$ be an additive category and $f$ a morphism in $\mathscr{C}$. We define:

1) the image of $f$ as the kernel of the cokernel of $f$, denoted by $\mathrm{Im}(f)$,
2) the coimage of $f$ as the cokernel of the kernel of $f$, denoted by $\mathrm{Coim}(f)$.

**Example III.4.**

For abelian groups, we recover the classical notions of image and coimage.

**Proposition III.1 ([Ass22], Proposition 4.3).**

Let $\mathscr{C}$ be an additive category in which every morphism admits a kernel and a cokernel. Let $f : X \to Y$ be a morphism.
Then, there exists a unique morphism $\bar{f} : \mathrm{Coim}(f) \to \mathrm{Im}(f)$ so that $f = j\bar{f}p$, meaning that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ker}(f) \arrow[r, "k"] & X \arrow[r, "f"] \arrow[d, two heads, "p"'] & Y \arrow[r, "c"] & \mathrm{Coker}(f) \\ & \mathrm{Coim}(f) \arrow[r, dashed, "\exists!\, \bar{f}"] & \mathrm{Im}(f) \arrow[u, hook, "j"'] & \end{tikzcd}$$

This decomposition is called the *canonical factorisation* of $f$.

### III.1.4 Abelian categories

See [Ass22, VIII §2, 3].

**Definition III.11.**

Let $\mathscr{C}$ be an additive category, so that every morphism admits kernel and cokernel, and $f$ a morphism.
$f$ is said to be *strict* if its canonical factorisation $\bar{f}$ (see III.1) is an isomorphism.

**Proposition III.2 ([Ass22], Theorem 2.8).**

With the same notation as the previous definition, the morphism $f$ is strict if, and only if there exist a kernel $j$ and a cokernel $p$ such that $f = jp$.

One can show that in this case, $j = \mathrm{Im}(f)$ and $p = \mathrm{Coim}(f)$ (see Corollary 2.10 in [Ass22]).

**Definition III.12.**

An additive category is said to be *abelian* if:

1. every morphism admits a kernel and a cokernel,
2. every morphism is strict.

Some special functors will be our main interest in the next sections.

**Definition III.13.**

Let $F : \mathscr{C} \to \mathscr{D}$ be a linear functor between abelian categories. Assume that $F$ is covariant (the contravariant case is dual).

1) $F$ is said to be *exact* if the exactness of

$$X \to Y \to Z$$

implies the one of

$$FX \to FY \to FZ$$

,

2) $F$ is said to be *left-exact* if the exactness of

$$0 \to X \to Y \to Z$$

implies the one of

$$0 \to FX \to FY \to FZ$$

,

3) $F$ is said to be *right-exact* if the exactness of

$$X \to Y \to Z \to 0$$

implies the one of

$$FX \to FY \to FZ \to 0.$$

### III.1.5 Homology, cohomology, homotopy

The categorical approach to (co)homology have historically been motivated by algebraic topology and geometry. It is entirely normal for the notation to remind the reader of concepts already encountered in applications.
We follow [Ass22, IX§5].
In this subsubsection, $\mathscr{C}$ is an abelian category.

#### III.1.5.a. Complexes of (co)chains

**Definition III.14.**

1) A complex of chains is the data of $C_\bullet = (C_n, d_n)_{n\in\mathbb{Z}}$ with the $C_n$'s objects of $\mathscr{C}$ and $d_n : C_n \to C_{n-1}$ such that for every $n \in \mathbb{Z}, d_n \circ d_{n+1} = 0$. We will denote:

$$C_\bullet : \dots \to C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \to \dots$$

2) A complex of cochains is the data of $C^\bullet = (C^n, d^n)_{n\in\mathbb{Z}}$ with the $C^n$'s objects of $\mathscr{C}$ and $d^n : C^n \to C^{n+1}$ such that for every $n \in \mathbb{Z}, d^n \circ d^{n-1} = 0$. We will denote:

$$C^\bullet : \dots \to C^{n-1} \xrightarrow{d^{n-1}} C^n \xrightarrow{d^n} C^{n+1} \to \dots$$

The rest of the text will mainly use the second definition, so we will state everything in this term, but the dual can easily be found by the reader.

**Definition III.15.**

We say that a complex of cochains $C^\bullet$ is *bounded above* (resp. *bounded below*) if there exists $i \in \mathbb{Z}$ such that for every $j \geq i$ (resp. $j \leq i$), $C^j = 0$.
$C^\bullet$ is said to be *bounded* if it is bounded below and above.

**Remark III.5.**

The condition $d^n \circ d^{n-1} = 0$ means $\mathrm{Im}(d^{n-1}) \subseteq \mathrm{Ker}(d^n)$ (when this notation makes sense in $\mathscr{C}$).

**Definition III.16.**

Let $C^\bullet$ be a complex of cochains.
$C^\bullet$ is said to be *exact* if for every $n \in \mathbb{Z}$ the following holds: $\mathrm{Ker}(d^n) = \mathrm{Im}(d^{n-1})$.

**Example III.6.**

Let $C^\bullet$ be a complex of cochains and $n \in \mathbb{Z}$. We say that $C^\bullet$ is *concentrated in degree $n$* if $C_m = 0$ for $n \neq m$. It is exact if, and only if, $C_m$ is also 0.

**Remark III.7.**

By abuse of notations, if the context if clear, we will denote all the applications between cochains $d^n$.

**Definition III.17.**

Let $C^\bullet$ and $D^\bullet$ two complexes of cochains.
A morphism $f^\bullet : C^\bullet \to C'^\bullet$ is a sequence of morphism $(f^n : C^n \to C'^n)_{n\in\mathbb{Z}}$ such that for every $n \in \mathbb{Z}, f^n \circ d^{n-1} = d'^n \circ f^{n-1}$, *ie* the following diagram commutes:

$$\begin{tikzcd}
\dots \arrow[r] & C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, "f^{n-1}"] & C^n \arrow[r, "d^n"] \arrow[d, "f^n"] & C^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] & \dots \\
\dots \arrow[r] & C'^{n-1} \arrow[r, "d'^{n-1}"] & C'^n \arrow[r, "d'^n"] & C'^{n+1} \arrow[r] & \dots .
\end{tikzcd}$$

With this construction, complexes of cochains (resp. bounded above, resp. bounded below, resp. bounded) form a category, denoted by $\mathrm{Comp}^\bullet(\mathscr{C})^\sharp$, with $\sharp = \emptyset, -, +, b$ respectively.

#### III.1.5.b. Cocycle, coboundary, cohomology

**Definition III.18.**

Let $C^\bullet$ be a complex of cochains.

1) The objects $\mathrm{Z}^n(C^\bullet) := \mathrm{Ker}(d^n)$ are called *$n$-cocycles* of the complex,
2) The objects $\mathrm{B}^n(C^\bullet) := \mathrm{Im}(d^{n-1})$ are called *$n$-coboundaries* of the complex,
3) The objects $\mathrm{H}^n(C^\bullet) := \mathrm{Z}^n(C^\bullet)/\mathrm{B}^n(C^\bullet)$ are called *$n$-cohomologies* of the complex.

**Lemma III.3.**

$\mathrm{Z}^n, \mathrm{B}^n$ and $\mathrm{H}^n$ induces covariant functors on $\mathrm{Comp}^\bullet(\mathscr{C})$, and there is an exact short sequence of functors:

$$0 \to \mathrm{B}^n \to \mathrm{Z}^n \to \mathrm{H}^n \to 0.$$

**Definition III.19.**

$\mathrm{H}^n$ is called the *$n$-th cohomology functor*.

**Definition III.20.**

A morphism $f^\bullet : C^\bullet \to D^\bullet$ is called a *quasi-isomorphism* if it induces an isomorphism in cohomology: for every $n \in \mathbb{Z}$,

$$\mathrm{H}^n(f^\bullet) : \mathrm{H}^n(C^\bullet) \xrightarrow{\sim} \mathrm{H}^n(D^\bullet).$$

The cohomology objects are a mean to measure the exactness default of a cochain at a certain degree: $\mathrm{H}^n(C^\bullet) = 0$ if, and only if $C^\bullet$ is exact at degree $n$.
The cohomology functor is not exact (meaning, it does not transform a short exact sequence in another one) in general, but gives rise to a *long exact sequence in cohomology*:

**Theorem III.4 ([Ass22], Theorem 5.12).**

Let $0 \to C'^\bullet \xrightarrow{f^\bullet} C^\bullet \xrightarrow{g^\bullet} C''^\bullet \to 0$ a short exact sequence of complexes of cochains. There exists an exact sequence:

$$\cdots \to \mathrm{H}^n(C'^\bullet) \xrightarrow{\mathrm{H}^n(f^\bullet)} \mathrm{H}^n(C^\bullet) \xrightarrow{\mathrm{H}^n(g^\bullet)} \mathrm{H}^n(C''^\bullet) \xrightarrow{\delta^n} \mathrm{H}^{n+1}(C'^\bullet) \xrightarrow{\mathrm{H}^{n+1}(f^\bullet)} \mathrm{H}^{n+1}(C'^\bullet) \to \dots$$

#### III.1.5.c. Homotopy

Two complexes of cochains can have the same $n$-th cohomology object (for all $n$) without being isomorphic, and two morphisms can induce the same map in the cohomology without being equal. An important relation between morphisms having inducing the same in cohomology is the *homotopy*:

**Definition III.21.**

Let $f^\bullet, g^\bullet : C^\bullet \to C'^\bullet$ two morphisms of complexes of cochains.
$f^\bullet$ and $g^\bullet$ are called *homotopic* if there exists a sequence $(s^n : C^n \to C'^{n-1})_{n\in\mathbb{Z}}$ such that for

all $n \in \mathbb{Z}$, $f^n - g^n = s^{n+1}d^n + d'^{n-1}s^n$:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, shift right, "g^{n-1}"'] \arrow[d, shift left, "f^{n-1}"] & C^n \arrow[r, "d^n"] \arrow[d, shift right, "g^n"'] \arrow[d, shift left, "f^n"] \arrow[dl, dashed, blue, "s^n"'] & C^{n+1} \arrow[r] \arrow[d, shift right, "g^{n+1}"'] \arrow[d, shift left, "f^{n+1}"] \arrow[dl, dashed, blue, "s^{n+1}"'] & \cdots \\
\cdots \arrow[r] & C'^{n-1} \arrow[r, "d'^{n-1}"'] & C'^n \arrow[r, "d'^n"'] & C'^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

**Lemma III.5 ([Ass22], Lemma 5.14).**

The homotopic relation on $\mathrm{Hom}_{\mathrm{Comp}^\bullet(\mathscr{C})}(C^\bullet, C'^\bullet)$ is an equivalence relation. Moreover, if $f^\bullet$ and $g^\bullet$ are homotopic, then $\mathrm{H}^n(f^\bullet) = \mathrm{H}^n(g^\bullet)$ for every $n \in \mathbb{Z}$.

### III.1.5.d. Double complexes and total complexes

See [[Ben22], III§5].
For the construction of the hypercohomology III.5.2.b, we will need a "resolution of the resolution": a double complex.

**Definition III.22.**

A *double cochain complex* $X^{\bullet\bullet}$ is a diagram of the form

$$\begin{tikzcd}
 & {} & {} & {} & \\
0 \arrow[r] & X^{(0,2)} \arrow[r, "d^h_{0,2}"] \arrow[u, "d^v_{0,2}"] & X^{(1,2)} \arrow[r, "d^h_{1,2}"] \arrow[u, "d^v_{1,2}"] & X^{(2,2)} \arrow[r, "d^h_{2,2}"] \arrow[u, "d^v_{2,2}"] & {} \\
0 \arrow[r] & X^{(0,1)} \arrow[r, "d^h_{0,1}"] \arrow[u, "d^v_{0,1}"] & X^{(1,1)} \arrow[r, "d^h_{1,1}"] \arrow[u, "d^v_{1,1}"] & X^{(2,1)} \arrow[r, "d^h_{2,1}"] \arrow[u, "d^v_{2,1}"] & {} \\
0 \arrow[r] & X^{(0,0)} \arrow[r, "d^h_{0,0}"] \arrow[u, "d^v_{0,0}"] & X^{(1,0)} \arrow[r, "d^h_{1,0}"] \arrow[u, "d^v_{1,0}"] & X^{(2,0)} \arrow[r, "d^h_{2,0}"] \arrow[u, "d^v_{2,0}"] & {} \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

where, for every pair $(i, j)$, we have

$$d^h_{i+1,j} \circ d^h_{i,j} = 0,\ d^v_{i,j+1} \circ d^v_{i,j} = 0,\ d^v_{i+1} \circ d^h_{i,j} = d^h_{i,j+1} \circ d^v_{i,j}.$$

**Definition III.23.**

Let $X^{\bullet\bullet}$ be a double cochain complex.
The *total complex* attached to $X^{\bullet\bullet}$ is the complex $\mathrm{Tot}(X^{\bullet\bullet}) = X^\bullet$, with

- $X^n := \bigoplus_{i+j=n} X^{i,j}$,
- $d_n = \sum_{i+j=n} d^h_{i,j} + (-1)^i d^v_{i,j} : X^n \to X^{n+1}$.

**Definition III.24.**

Let $Y^\bullet$ a cochain complex and $X^{\bullet\bullet}$ a double cochain complex.
$X^{\bullet\bullet}$ is said to be an *exact resolution* of $Y^\bullet$ if there exist maps $u^i : Y^i \to X^{(i,0)}$ such that

$$\begin{tikzcd}
 & {} & {} & {} & \\
0 \arrow[r] & X^{(0,1)} \arrow[u, "d^v_{0,1}"] \arrow[r, "d^h_{0,1}"] & X^{(1,1)} \arrow[u, "d^v_{1,1}"] \arrow[r, "d^h_{1,1}"] & X^{(2,1)} \arrow[u, "d^v_{2,1}"] \arrow[r, "d^h_{2,1}"] & {} \\
0 \arrow[r] & X^{(0,0)} \arrow[u, "d^v_{0,0}"] \arrow[r, "d^h_{0,0}"] & X^{(1,0)} \arrow[u, "d^v_{1,0}"] \arrow[r, "d^h_{1,0}"] & X^{(2,0)} \arrow[u, "d^v_{2,0}"] \arrow[r, "d^h_{2,0}"] & {} \\
0 \arrow[r] & Y^0 \arrow[u, "u^0"] \arrow[r, "d^0"] & Y^1 \arrow[u, "u^1"] \arrow[r, "d^1"] & Y^2 \arrow[u, "u^2"] \arrow[r, "d^2"] & {} \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

commutes, and has exact columns.

### III.1.6 Derived categories

Our use of the derived categories will be superficial in this text: they will only appear in VI, especially since we will not give proofs.
We refer the curious reader to the presentation in [[HTT08], Appendix B], in which are being given references for more detailed construction: here is only a very brief description.

#### III.1.6.a. Derived categories

Starting an abelian category $\mathscr{C}$, we define its $\sharp$-homotopy category as its category of $\sharp$-complexes quotiented out by the homotopy equivalence (with again $\sharp = \emptyset, -, +, b$).
It is not an abelian category anymore, so the theory of *triangluated triangle* is to be developped. To get the $\sharp$-derived category $D^\sharp(\mathscr{C})$, we make invertible quasi-isomorphisms by a *localization process* (similar to the one in ring theory).
When we work with derived categories, we will think of there objects as chain of complexes of cochains with some relations.

#### III.1.6.b. Derived functors

What will be more useful are *derived functors*, essential in the cohomological theory. We are sketching here the construction, following [[Ben22], Chapter 3].
The goal is to approximate a right(or left)-exact functor by an exact functor. Some tools of exactness could be then used. They are also the natural functors to consider when working with derived

categories.
In this subsubsection, $\mathscr{C}$ is an abelian category.

**Definition III.25.**

1) a. An object $P$ of $\mathscr{C}$ is said to be *projective* if $\mathrm{Hom}_{\mathscr{C}}(P,-)$ is an exact functor.
   b. $\mathscr{C}$ is said to *have enough projective objects* if for every $X \in \mathrm{Ob}(\mathscr{C})$, there exists a projective object $P$ and an epimorphism $f : P \to X$.
2) a. An object $I$ of $\mathscr{C}$ is said to be *injective* if $\mathrm{Hom}_{\mathscr{C}}(-,I)$ is an exact functor.
   b. $\mathscr{C}$ is said to *have enough injective objects* if for every $X \in \mathrm{Ob}(\mathscr{C})$, there exists a injective object $I$ and a monomorphism $g : X \to I$.

As these two notions are dual, and that the injective aspect proves more useful in the cohomological setup later on, the remainder will focus on the latter.

**Definition III.26.**

Let $X \in \mathrm{Ob}(\mathscr{C})$.
A *right resolution* of $X$ is a sequence $I^\bullet : \dots I^0 \to I^1 \to \dots$ equipped with a morphism $e : X \to I^0$ such that

$$0 \to X \xrightarrow{e} I^0 \to I^1 \to \dots$$

is an exact sequence.
If all the $I$'s are injective, the resolution is said to be *injective.*

By induction, we have the following result:

**Proposition III.6.**

Assume that $\mathscr{C}$ has enough injectives.
Then every object of $\mathscr{C}$ has an injective resolution.

Let $F : \mathscr{C} \to \mathscr{D}$ henceforth be a covariant left exact functor, with $\mathscr{C}$ and $\mathscr{D}$ abelian, and assume that $\mathscr{C}$ has enough injectives.
We construct the right exact functor associated to $F$ and describe some of its properties. First, a lemma that is useful to the proofs (and helps grasping an intuition behind them):

**Lemma III.7.**

Let $X$ and $Y$ objects of $\mathscr{C}$, let $I^\bullet$ an injective resolution of $X$, $J^\bullet$ a (arbitrary) right resolution of $Y$, and let $f : X \to Y$.
Then, there exists a morphism of chains $f^\bullet : I^\bullet \to J^\bullet$, unique up to homotopy, such that the diagram commutes

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r] \arrow[d, "f"] & I^0 \arrow[r] \arrow[d, "f^0"] & I^1 \arrow[r] \arrow[d, "f^1"] & \dots \\
0 \arrow[r] & Y \arrow[r] & J^0 \arrow[r] & J^1 \arrow[r] & \dots
\end{tikzcd}$$

We can apply the functor $F$ to $I^\bullet$, but the sequence obtained $0 \to F(X) \to F(I^0) \to \dots$ is not exact in general since $F$ is only exact on the left.

**Proposition III.8.**

Let $X \in \mathrm{Ob}(\mathscr{C})$ and $I^\bullet, J^\bullet$ two injective resolutions of $X$.
Then for every $n \in \mathbb{N},\ \mathrm{H}^n(F(I^\bullet)) \simeq \mathrm{H}^n(F(J^\bullet))$.

**Definition III.27.**

For every $n \in \mathbb{N}$, we define

$$\mathrm{R}^n F(X) := \mathrm{H}^n(F(I^\bullet))$$

for any injective resolution $I^\bullet$ of $X$.

Here is a list of some properties of this object:

**Proposition III.9.**

Let $X$ be an object of $\mathscr{C}$.

1) $\mathrm{R}^0 F(X) \simeq F(X)$,
2) $\mathrm{R}^n F(X)$ is well defined, up to canonical isomorphisms,
3) If $X$ is injective, for every $n \geq 1, \mathrm{R}^n F(X) = 0$,
4) Each morphism $f : X \to Y$ induces a canonical $\mathrm{R}^n F(f) : \mathrm{R}^n F(X) \to \mathrm{R}^n F(Y)$ for every $n \geq 0$,
5) For every $n \geq 0, \mathrm{R}^n F(-) : \mathscr{C} \to \mathscr{D}$ is an additive functor and $\mathrm{R}^0 F \simeq F$.
6) Let $0 \to X' \to X \to X'' \to 0$ a short exact sequence. There exists a long exact sequence in the derived world:

$$0 \to \mathrm{R}^0 F(X') \to \mathrm{R}^0 F(X) \to \mathrm{R}^0 F(X'') \to \mathrm{R}^1 F(X') \to \dots.$$

**Definition III.28.**

The functors defined in 5) are called the *nth right derived functors* of $F$.

A injective resolution is usually hard to find. An other class of objects that can be used to compute the derived functors are acyclic objects:

**Definition III.29.**

An object $J$ of $\mathscr{C}$ is said to be *acyclic* if for every $n \geq 1,\ \mathrm{R}^n F(J) = 0$, and a resolution is said to be acyclic if all its objects are acyclic.

**Proposition III.10.**

Let $J^\bullet$ an acyclic resolution of $X$.
Then for every $n \in \mathbb{N},\ \mathrm{R}^n F(X) \simeq \mathrm{H}^n(F(J^\bullet))$.

**Remark III.8.**

Why are acyclic resolutions useful to compute derived functors if they are build satifying a condition on the said derived functors?

> The answer is that sheaf-theoric properties on some resolutions can give the acyclicity condition, as *flabby sheaves* [[Voi02], Definition 4.33] or *fine sheaves* [[Voi02], Definition 4.35].

Back to double complexes, we have the following:

**Proposition III.11.**

Assume that all the columns (respectively all raws) of $X^{\bullet\bullet}$ are exact.
Then, $\mathrm{Tot}(X^{\bullet\bullet})$ is acyclic.

Examples of such derived functors can be found in III.5.

## III.2 Sheaf theory

This section will contain more examples because I gave a talk on the subject in the $\mathcal{D}$-seminar.
A classic reference is [[Har77], II§1].
In the following, $X$ is a topological space and $\mathcal{T}_X$ will denote its topology category.

### III.2.1 First definitions

**Definition III.30.**

Let $\mathscr{C}$ be a category (often $\mathscr{C} = \mathscr{S}et, \mathscr{A}b, \mathscr{R}ings...$).
A *presheaf* of $\mathscr{C}$ is a contravariant functor $\mathcal{F} : \mathcal{T}_X \to \mathscr{C}$.
Explicitly, it is the data of:

1. for every open set $U$ of $X$, an object $\mathcal{F}(U)$ of $\mathscr{C}$,
2. for every couple of open sets $V \subseteq U$ of $X$, a morphism $\mathrm{res}_{U,V} : \mathcal{F}(V) \to \mathcal{F}(U)$ verifying:
   a. for every open set $U$ of $X$, then $\mathrm{res}_{U,U} = \mathrm{id}_U$,
   b. for every open sets $U \subseteq V \subseteq W$ of $X$, then $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

The objects $\mathcal{F}(U)$ are often called *sections of $\mathcal{F}$ over $U$*, and for avoiding confusion, will be denoted $\Gamma(U, \mathcal{F})$.

**Remark III.9.**

If the context is clear, we will denote $\mathrm{res}_{U,V}(f) = f|_U$ (the notation is justified in the next lines).

**Example III.10.**

1) The sheaf of real valued continuous functions $C^0(-, \mathbb{R})$ with classical restriction functions $\mathrm{res}(U, V)(f) = f|_U$.
2) If $X$ has an additional structure, the above example can be further specified: if $X$ is a differentiable manifold, we can consider $C^\infty(-, \mathbb{R})$, if $X$ is a complex manifold, $\mathscr{O}(-, \mathbb{C})$ ...
3) If $C$ is an object of $\mathscr{C}$, the constant presheaf $\underline{C}$ or $C_X$. We will be intersted in $\mathbb{Z}_X$ or $\mathbb{C}_X$ for example.
4) The sheaf of real valued bounded functions $B(-, \mathbb{R})$.

Presheaves comes from a category point of view, so we need morphisms:

**Definition III.31.**

A *morphism of presheaves* (over the same space $X$) $\varphi : \mathcal{F} \to \mathcal{G}$ is a natural transformation. More explicitly, it is the data, for each $U \in \mathcal{T}_X$, of a morphism $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ of $\mathscr{C}$ such that the following diagram commutes for every $U \subseteq V$:

$$\begin{tikzcd} \mathcal{F}(V) \arrow[r, "\varphi_V"] \arrow[d, "\mathrm{res}^{\mathcal{F}}_{U,V}"'] & \mathcal{G}(V) \arrow[d, "\mathrm{res}^{\mathcal{G}}_{U,V}"] \\ \mathcal{F}(U) \arrow[r, "\varphi_U"] & \mathcal{G}(U) \end{tikzcd}$$

**Definition III.32.**

A *sheaf* $\mathcal{F}$ is a presheaf satisfying the extra conditions for any open cover $\{U_i, i \in I\}$ of an open set $U$ of $X$:

3. for every $f, g \in \mathcal{F}(U)$ such that: $\forall i \in I,\ f|_{U_i} = g|_{U_i}$, then $f = g$.
4. for every $(f_i) \in \prod_{i \in I} \mathcal{F}(U_i)$ such that: $\forall (i,j) \in I^2,\ f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$, then there exists $f \in \mathcal{F}(U)$ with $f|_{U_i} = f_i$ for any $i \in I$.

**Remark III.11.**

For $U = \emptyset$, it says that $\mathcal{F}(\emptyset)$ is the terminal object of $\mathscr{C}$ (if such one exists).

**Example III.12.**

- 1) and 2) are indeed sheaves.
- 3) is not, because of the previous remark.
- 4) is not either in general, with the conterexample $\mathbb{R} = \bigcup_{n \in \mathbb{N}} (-n, n)$ and $f : x \mapsto x$.
- If $X$ is Hausdorff, $x_*$ is a point of $X$, $(S, s)$ a pointed set, the *skyscraper sheaf* is given by $\mathrm{sky}_{x_*}(S) : U \mapsto \begin{cases} S & \text{if } x_* \in U \\ s & \text{otherwise.} \end{cases}$

**Definition III.33.**

We will denote by $\mathrm{PSh}_X$, respectively $\mathrm{Sh}_X$, the category of presheaves, respectively the full subcategory of sheaves, over $X$ (implicitly taking values in a certain category).

### III.2.2 The local point of view

In the following, we assume that $\mathscr{C}$ is the category of abelian groups. One can ask what looks like the behavior of the sheaf very locally, on a point. Since a singleton is in general not an open, we need the following definition.

**Definition III.34.**

Let $\mathcal{F}$ be a (pre)sheaf.
We define the *stalk* at $x \in X$ as:

$$\mathcal{F}_x := \{(U, s) |\ x \in U,\ U \in \mathcal{T}_X,\ s \in \mathcal{F}(U)\} \big/ \sim\ ,$$

where $(U,s) \sim (V,t) \Leftrightarrow (\exists W \subset U \cap V \text{ open}) : s|_W = t|_W$.

**Remark III.13.**

From a categorical point of view, it is also the direct limit: $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$, where the limit is taken over open set $U$ containing $x$.

### III.2.3 Sheafification

It turns out that this localization principle can be used to transform a presheaf into a sheaf in a natural way (in the sense of what follows).

**Definition III.35.**

Let $\mathcal{F}$ be a presheaf.
We define for every open set $U$, the set $\mathcal{F}^\sharp(U)$ to be:

$$\left\{ s : U \to \coprod_{x \in U} \mathcal{F}_x \ \middle| \ \left\{ \begin{array}{l} \forall x \in U,\ s(x) \in \mathcal{F}_x \\ \exists V \subset U, \exists t \in \mathcal{F}(V) \text{ such that } x \in V \text{ and } \forall y \in V,\ s(y) = t(y) \end{array} \right. \right\}.$$

$\mathcal{F}^\sharp$ is called the *sheafification of* $\mathcal{F}$.

**Proposition III.12 ([Har77], II§1, Proposition 1.2).**

$\mathcal{F}^\sharp$ is a sheaf.

**Example III.14.**

$\underline{\mathbb{R}}^\sharp$ is the sheaf of locally constant functions, more precisely the sheaf of continuous functions $C^0(-,\mathbb{R})$, where $\mathbb{R}$ is equipped with the discrete topology.

The proof in the Harsthone is based on a universal property, that can be reformulated this way:

**Proposition III.13.**

$((-)^\sharp, \text{Forget})$ is an adjoint pair of functors between $\mathrm{PSh}_X$ and $\mathrm{Sh}_X$.

**Example III.15.**

Suppose that $\mathscr{C}$ admits kernels and cokernels (this assumption holds whenever kernels and cokernels are written).
For $\varphi : \mathcal{F} \to \mathcal{G}$, we define $\mathrm{Ker}(\varphi)(U) := \mathrm{Ker}(\varphi_U)$, $\mathrm{Im}(\varphi)(U) := \mathrm{Im}(\varphi_U)$ and $\mathrm{Coker}(\varphi)(U) := \mathrm{Coker}(\varphi_U)$.
The first one is always a sheaf, but the two others are not in general. In the following, $\mathrm{Im}(\varphi)$ or $\mathrm{Coker}(\varphi)$ will always denote the sheafification of these presheaves.

**Proposition III.14 ([Har77], Ex. 1.2).**

A complex $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is exact if, and only if the induced complex on stalks

$0 \to \mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x \to 0$ is exact for every $x \in X$.

**Remark III.16.**

In general, it is not equivalent of the exactness of the induced sequence over every open set $U$. One could take as a counterexample the exponential sequence on $\mathbb{C}$:

$$\begin{array}{ccccccccc} 0 & \to & \underline{\mathbb{Z}} & \to & \mathscr{O}_{\mathbb{C}} & \to & \mathscr{O}^*_{\mathbb{C}} & \to & 0, \\ & & & & f & \mapsto & \exp(2\pi i f) & & \end{array}$$

that is not exact over $\mathbb{C}^*$, because the map $\Gamma(\mathbb{C}^*, \mathscr{O}_{\mathbb{C}^*}) \to \Gamma(\mathbb{C}^*, \mathscr{O}^*_{\mathbb{C}})$ is not surjective: the map $z \mapsto z$ does not belong to the image since log is not well defined over $\mathbb{C}^*$.
This default in the exactness of these short sequences actually gives rise to the important theory of *sheaf cohomology*, see III.5.1.

### III.2.4 Some useful general tools

The following construction will be useful in the next sections.

**Proposition III.15 ([Sza12], Construction 2.7.8).**

Let $X = \cup_{i \in I} U_i$ an open covering of $X$. For every $i \in I$, let $\mathcal{F}_i$ a sheaf on $U_i$.
Assume that for every $i \neq j \in I$, we have chosen an isomorphism

$$\theta_{i,j} : \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j},$$

satisfying the condition $\theta_{j,k} \circ \theta_{i,j} = \theta_{i,k}$ on $U_i \cap U_j \cap U_k$ for every triplet of distinct elements of $I$.
Then, there exists a sheaf $\mathcal{F}$ on $X$ such that for every $i \in I, \mathcal{F}|_{U_i} = \mathcal{F}_i$. Moreover, this sheaf is unique, up to unique isomorphism.

For a continuous map $f : X \to Y$ (with something more assumptions on $f$, $X$ or $Y$), there exists numerous functors assocated on the sheaves: $f_*, f^*, f_!$ etc. We present the "easier" one to define:

**Definition III.36.**

Let $f : X \to Y$ a continuous map.
The induced functor

$$f_* : \mathrm{Sh}_X \to \mathrm{Sh}_Y,$$

$$f_* F(U) = F(f^{-1}(U)),$$

is called the *direct image functor*.

**Definition III.37.**

A *ringed space* is a pair $(X, \mathcal{O}_X)$[a], with $X$ a topological space and $\mathcal{O}_X$ a sheaf of ring over $X$, often called the *structural sheaf* of the ringed space.

[a]the notation is justify in III.4

**Definition III.38.**

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ two ringed spaces.
A *morphism of ringed spaces* $(f, \phi) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a pair constisting of a continuous map $\phi : X \to Y$ and a morphism of sheaves $\phi : \mathcal{O}_Y \to f_*\mathcal{O}_X$.

**Definition III.39.**

Let $(X, \mathcal{O}_X)$ be a ringed space.
An $\mathcal{O}_X$*-module* $M$ is a sheaf on $X$ such that, for every open set $U$ of $X$, $M(U)$ has the structure of a $\mathcal{O}_X(U)$-module, and the following diagram commutes for $U \subset V$:

$$\begin{array}{ccc} \mathcal{O}_X(V) \times M(V) & \longrightarrow & M(U) \\ \Big\downarrow \mathrm{res}_{U,V} & & \Big\downarrow \mathrm{res}_{U,V} \\ \mathcal{O}_X(U) \times M(U) & \longrightarrow & M(U), \end{array}$$

where the horizontal arrows are given by the multiplications.

**Definition III.40.**

An $\mathcal{O}_X$-module $M$ is said to be *locally free or rank* $r$ $(r \in \mathbb{N})$ if, for every $x \in X$, there exists an open neighbourhood $U$ of $x$ such that:

$$M|_U \simeq (\mathcal{O}_X|_U)^{\oplus r}.$$

We denote by $\mathrm{LocFree}_r(X)$ the category of locally free $\mathcal{O}_X$-modules or rank $r$.

## III.3 Complex geometry

A good reference for this subsecion is [Huy04].

### III.3.1 First definitions

See [[Huy04], 2.1].

**Definition III.41.**

Let $X$ be a differentiable manifold of dimension $2n$.
A *holomorphic atlas* is a set $\{(U_i, \varphi_i), i \in I\}$ where $X = \bigcup_{i \in I} U_i$ is an open cover and $\varphi : U_i \to \mathbb{C}^n$ are homeomorphisms on their image, such that for all $(i,j) \in I^2$, the *transition map* $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ is holomorphic.

Such a couple $(U_i, \varphi_i)$ is called a *local coordinate system*, or just *coordinates*.

We say that two holomorpic atlases $\{(U_i, \varphi_i), i \in I\}$ and $\{(\tilde{U}_j, \tilde{\varphi}_j), j \in J\}$ are *equivalent* if for all $(i,j) \in I \times J, \tilde{\varphi}_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap \tilde{U}_j) \to \tilde{\varphi}_j(U_i \cap \tilde{U}_j)$.

**Definition III.42.**

A *complex manifold of dimension* $n$ is a real manifold $X$ of dimension $2n$ endowed with an equivalence class of holomorphic atlases.

**Remark III.17.**

In the following, unless explicitly stated otherwise, the complex manifold will be endowed with a representative of the equivalence class, denoted by $\{(U_i, \varphi_i), i \in I\}$.

**Definition III.43.**

Let $X$ be a complex manifold of dimension $n$ and $Y \subset X$ be a differentiable manifold of real dimension $2k$.
$Y$ is said to be a *complex submanifold of dimension* $k$ of $X$ if there exists a holomorphic atlas $\{(U_i, \varphi_i), i \in I\}$ of $X$ such that $\varphi_i : U_i \cap Y \xrightarrow{\sim} \varphi_i(U_i) \cap \mathbb{C}^k$.
The number $n-k$ is called the *codimension* of $Y$ in $X$.

**Definition III.44.**

Let $X$ be a complex manifold and $f : X \to \mathbb{C}$.
We say that $f$ is *holomorphic* if for all local coordinates $(U, \varphi)$, $f \circ \varphi : \varphi(U) \to \mathbb{C}$ is holomorphic.

**Definition III.45.**

Let $X$ be a complex manifold.
We denote by $\mathscr{O}_X$ the sheaf of holomorphic function on $X$, *ie*:
$\mathscr{O}_X : U \mapsto \{f : U \to \mathbb{C}, f \text{ holomorphic}\}$.

**Remark III.18.**

$\mathscr{O}_{X,x}$ will denote the stalk of $\mathscr{O}_X$ at $x$. It coincides with the germs of holomorphic functions at $x$. *Via* local coordinates $(U, \varphi)$, with $\varphi(x_*) = 0$ for some $x_* \in U$, we have an isomorphism $\mathscr{O}_{X,x} \simeq \mathscr{O}_{\mathbb{C}^n,0}$. It is an integral domain, and we denote by $Q(\mathscr{O}_{X,x})$ the associated fraction field.

More generally, if the codomain is also a complex manifold, we have the following definition:

**Definition III.46.**

Let $X$ and $Y$ be complex manifolds and $f : X \to Y$ continuous.
We say that $f$ is *holomorphic* if for all local coordinates $(U, \varphi)$ on $X$ and $(U', \varphi')$ on $Y$, $\varphi' \circ f \circ \varphi : \varphi(U \cap f^{-1}(U')) \to \varphi'(U')$ is holomorphic.

As for a differentiable manifold, one can define holomorphic vector field and holomorphic $p$-forms. We will denote them $\Theta_X$ and $\Omega^p_X = \bigwedge^p \Omega^1_X$ respectively.

**Definition III.47.**

Let $X$ be a complex manifold.
A *meromorphic function* on $X$ is a map $f : X \to \bigsqcup_{x \in X} Q(\mathscr{O}_{X,x})$ so that, for all $x_* \in X$, there exists an open neighborhood $U \subset X$ of $x_*$ and $g, h \in \mathscr{O}_X(U)$ verifying: $(\forall x \in U),\ f_x = \frac{g}{h}$.

The sheaf of meromorphic functions is denoted $\mathcal{K}_X$.

**Remark III.19.**

If $U \subset X$ is a connected open subset, then $\mathcal{K}_X(U)$ is a field, and if $X$ itself is connected, $\mathcal{K}(X) = \mathcal{K}_X(X)$ is called the function field of $X$.

### III.3.2 Vector bundles

See [[Huy04], 2.2].

**Definition III.48.**

Let $X$ be a complex manifold and $r$ a positive integer.
A *holomorphic vector bundle* or rank $r$ on $X$ is a complex manifold $E$ equipped with a holomorphic map $\pi : E \to X$ such that:

1. for all $p \in X$, the *fiber* $E_p := \pi^{-1}(p)$ can be equipped with a structure of complex vector space of dimension $r$,
2. there exists an open covering $X = \bigcup_{i \in I} U_i$ and biholomorphic maps $\psi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{C}^r$, called the *trivialization maps*, satisfying:
   a. the following diagram commutes:

$$\begin{tikzcd} \pi^{-1}(U_i) \arrow[rr, "\psi_i"] \arrow[dr, "\pi"'] & & U_i \times \mathbb{C}^r \arrow[dl, "\mathrm{pr}_1"] \\ & U_i & \end{tikzcd}$$

   b. The induced map $E_p \to \mathbb{C}^r$ for any $p \in U_i$ is $\mathbb{C}$-linear.

If $r = 1$, we say that $(E, \pi)$ is a *line bundle*.

**Remark III.20.**

The trivialization maps induce holomorphic maps $\psi_{i,j} = \psi_i \circ \psi_j^{-1} : U_i \cap U_j \to \mathrm{GL}_r(\mathbb{C})$, and the collection $\{\psi_{i,j}, i, j \in I\}$ is called a cocycle.
In particular, it verifies the conditions:

i. $\psi_{i,i} = \mathrm{id}_{U_i}$,
ii. $\psi_{j,k} \circ \psi_{i,j} = \psi_{i,k}$.

**Definition III.49.**

Let $(E_1, \pi_1)$ and $(E_2, \pi_2)$ two vector bundles on a complex vector $X$.
A map $\phi : (E_1, \pi_1) \to (E_2, \pi_2)$ is a holomorphic map $\phi : E_1 \to E_2$ satifying $\pi_2 \circ \phi = \pi_1$.

We denote by $\mathrm{VectBun}_r(X)$ the category of vector bundles or rank $r$ over $X$.

**Proposition III.16 ([Huy04], Proposition 2.2.19).**

Let $X$ be a complex manifold and let $r$ be a positive integer.
The categories $\mathrm{VectBun}^r(X)$ and $\mathrm{LocFree}_r(X)$ are equivalent.

### III.3.3 Connections

See [[Huy04], 4.2].

**Definition III.50.**

Let $\mathcal{E}$ be a locally free $\mathscr{O}_X$-module.
A *connection on* $\mathcal{E}$ is a $\mathbb{C}$-linear map $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ satisfying the *Leibniz relation*: for any local section $s$ and local function $f$, then

$$\nabla(fs) = df \otimes s + f\nabla(s).$$

In particular, it induces, for each integer $k > 0$, a $\mathbb{C}$-linear map

$$\begin{array}{rcl} \nabla^{(k)}: & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^k_X & \longrightarrow \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^{k+1}_X \\ & s \otimes \omega & \longmapsto (-1)^k \nabla s \wedge \omega + s \otimes dw \end{array}$$

A connection $\nabla$ is said to be *flat* (or *integrable*) if the *curvature* $K := \nabla^{(1)} \circ \nabla$ is zero.

**Remark III.21.**

In dimension 1, we have $\Omega^2_X = 0$, so all connections are flat.

**Example III.22.**

1) $d$ is a connection on $\mathscr{O}_X$ by identifying $\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \Omega^1_X$. It induces a connection on $\mathscr{O}_X^r$ acting coordinate-wise.
2) For every $\omega \in \Omega^1_X$, the map

$$\begin{array}{rcl} d + \omega : & \mathscr{O}_X & \longrightarrow \Omega^1_X \\ & f & \longmapsto df + \omega f \end{array}$$

is a connection. More generally, if $\omega$ is a, $r \times r$ matrix with coefficients in $\Omega^1_X$, the map $d + \omega$ is a connection on $\mathscr{O}_X^r$.

The following result says that these are the only ones on $\mathscr{O}_X^r$, and then the only ones locally on $\mathcal{E}$.

**Lemma III.17 ([Poo25], Proposition 7.4).**

Every connection $\nabla$ on $\mathscr{O}_X^r$ is written $d + \omega$ for some $\omega$ as above.

Proof

For every $f \in \mathscr{O}_X$ and $h \in \mathscr{O}_X^r$, we have by the Leibniz rule:

$$(\nabla - d)(fh) = f\nabla(h) + df \otimes h - fdh - df \otimes h = f(\nabla - d)h.$$

Thus, $\nabla - d : \mathscr{O}_X^r \to \Omega^1_X$ is $\mathscr{O}_X$-linear and it's the multiplication by a matrix $\omega$.

□

**Remark III.23.**

Connections are a reformulation of (systems) of differential equations, locally on a vector bundle.

Let $\nabla = d + \omega$ a connection on $\mathscr{O}_X^r$ (it is locally of this form by the previous lemma). We choose local coordinates $z^1, \ldots, z^n$ on $X$. Then we can write

$$\omega(z) = \sum_{j=1}^{n} A_i(z) dz^j,$$

where the $A_i$'s are matrices with holomorphic entries.
For a $f$ holomorphic on the chart, we then have:

$$\nabla f = \sum_{j=1}^{n} \left( \frac{\partial f}{\partial z^j} + A_j f \right) dz^j,$$

and $\nabla f = 0$ is equivalent to:

$$\begin{cases} \dfrac{\partial f}{\partial z^1}(z) = -A_1(z) f(z), \\ \qquad\qquad \vdots \\ \dfrac{\partial f}{\partial z^n}(z) = -A_n(z) f(z), \end{cases}$$

which is a system of *partial* differential equations.

**Definition III.51.**

We define the category $\mathrm{LocFree}^{\nabla}(X)$ with

- locally free $\mathscr{O}_X$-modules equipped with flat connections as objects,
- morphisms of $\mathscr{O}_X$-modules $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ making

$$\begin{tikzcd} \mathcal{E}_1 \arrow[r, "\nabla_1"] \arrow[d, "\varphi"] & \mathcal{E}_1 \otimes_{\mathscr{O}_X} \Omega_X^1 \arrow[d, "\varphi \otimes 1"] \\ \mathcal{E}_2 \arrow[r, "\nabla_2"] & \mathcal{E}_2 \otimes_{\mathscr{O}_X} \Omega_X^1 \end{tikzcd}$$

commutes, as morphisms between $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$.

Here are some algebraic constructions one can obtain with connections:

**Proposition III.18.**

Let $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$ two connections on $X$. The following are also connections:

1) $(\mathcal{E}_1^*, (- \otimes 1) \circ \nabla_1)$,
2) $(\mathcal{E}_1 \oplus \mathcal{E}_2, \nabla_1 \oplus \nabla_2)$,
3) $(\mathcal{E}_1 \otimes_{\mathscr{O}_X} \mathcal{E}_2, \nabla_1 \otimes 1 + 1 \otimes \nabla_2)$
4) $(\mathcal{H}om_{\mathscr{O}_X}(\mathcal{E}_1, \mathcal{E}_2), \phi \mapsto (\nabla_2 \circ \phi) - (\phi \otimes 1) \circ \nabla_1)$

### III.3.4 Complex analytic space

"Analytic variety" is a new step towards algebraic geometry. It allows "singularities" not seen be complex manifold.

**Definition III.52.**

Let $(X, \mathscr{O}_X)$ be a topological space $X$ endowed with a sheaf of $\mathbb{C}$-algebra.
We say that $(X, \mathscr{O}_X)$ is a *complex analytic space* if for every $x \in X$, there exists an open neighbourhood $U$ of $x$, an open $V$ of $\mathbb{C}^n$ for some $m$ and some holomorphic $f_1, \dots, f_m$ defined on $V$, such that $(U, \mathscr{O}_U)$ is isomorphic as a ringed space to $(Y, (\mathscr{O}_V/(f_1, \dots, f_m))|_Y)$ for $Y = \{x \in V,\ f_1(x) = \cdots = f_m(x) = 0\}$.

The following definition is fundamental in the statements of the Riemann-Hilbert correspondence: it gives a more general point of view for "union of hypersurfaces".

**Definition III.53.**

Let $X$ be a complex manifold.
A *divisor* on $X$ is a finite formal sum $\sum a_i X_i$ with integer coefficients $a_i$ of hypersurfaces (subvariety of codimension 1) $X_i$.

These definitions suffice in the rest of the report. The general theory is quite similar to the algebraic geometry one developed in the next section, see for example [[Huy04], Definition 1.1.23 and below].

## III.4 Algebraic geometry

### III.4.1 Algebraic sets, regular functions

The point of view of algebraic varieties suffices here. We only treat the affine case. A good reference is [Per08], chapter 1 and 3.
In this subsection, $k$ is a field.

**Definition III.54.**

Let $S \subset k[X_1, \dots, X_n]$.
The *affine algebraic set* defined by $S$ is the set $V(S) = \{\underline{x} \in k^n \mid \forall P \in S, P(\underline{x}) = 0\}$.

**Definition III.55.**

The *Zariski topology* on $k^n$ is the topology whose the closed sets are affine algebraic sets.

**Definition III.56.**

Let $V$ be an affine algebraic set of $k^n$.

1) The *ideal* attached to $V$ is the set

$$I(V) = \{P \in k[X_1, \dots, X_n] \mid \forall \underline{x} \in V, P(\underline{x}) = 0\}.$$

2) The *affine algebra* attached to $V$ is the set

$$\Gamma(V) := k[X_1, \dots, X_n]/I(V).$$

For a element $f \in \Gamma(V)$, we denote by $D(f)$ the non-vanishing locus of $f$: $D(f) = \{\underline{x} \in k^n \mid f(\underline{x}) = 0\}$.

**Proposition III.19 ([Per08], Definition 2.3 and above).**

Let $V$ be an affine algebraic set.
There exists a sheaf of rings over $V$, denoted by $\mathcal{O}_V$ and called the *sheaf of regular functions over* $V$, such that for every $f \in \Gamma(V)$:

$$\Gamma(D(f), \mathcal{O}_V) = \Gamma(V)_f,$$

where $\Gamma$ on the left means sections over $D(f)$, and $\Gamma(V)_f$ on the right the localized of $\Gamma(V)$ at $f$.

### III.4.2 Algebraic varieties

For a affine variety $V$, the space $(V, \mathcal{O}_V)$ is a ringed space, and a morphism between such spaces is given by the pre-composition $g \mapsto g \circ f$.

**Definition III.57.**

An *algebraic affine variety* is a ringed space $(X, \mathcal{O}_X)$, which is isomorphic (as ringed space) to an affine variety equipped with its sheaf of regular functions $(V, \mathcal{O}_V)$.

**Definition III.58 ([Per08], Definition III.4.1).**

An *algebraic variety* is a quasi-compact ringed space which is locally isomorphic to an algebraic affine variety.
A map between algebraic varieties is simply a map of ringed spaces.

### III.4.3 Irreducibility, dimension

**Definition III.59.**

Let $V$ be an affine algebraic space.
$V$ is said to be *irreducible* if for every closed set $F, G$ of $V$ such that $V = F \cup G$, then $V \in \{F, G\}$.

Let us assume our algebraic varieties to be irreducible in the rest of the report.
This property can be re-interpreted in algebraic terms of $I(V)$ or $\Gamma(V)$, see [[Per08], Theorem 3.2].
Now we express the notion of dimension, as for a complex manifold for example.

**Definition III.60.**

Let $X$ be an algebraic variety.
The dimension of $X$ is the maximum length of chains $X_1 \subsetneq X_2 \subsetneq \cdots \subsetneq X_n$ of irreducible closed sets. It is denoted by $\dim(X)$.

There exists a lot of ways to compute the dimension of an algebraic variety, more algebraic for example, see [[Per08], Chapter IV].

### III.4.4 Smoothness

**Definition III.61.**

Let A be a $k$-algebra and $M$ be an $A$-module.
A map $D : A \to M$ is called a *derivation* if it satisfies the following conditions:

1. $D$ is $k$-linear,
2. For every $a, b \in A$, $D(ab) = aD(b) + bD(a)$.

**Definition III.62.**

Let $X$ be an algebraic variety and $x \in X$.
The *tangent space at $x$* is the set of all derivation from $\Gamma(X)$ to $k$ (with the inherited action $f.v := f(x)v$). In other words, it is the set:

$$\mathrm{T}_x X := \{v : \Gamma(X) \to k \text{ linear} \mid \forall f, g \in \Gamma(X), v(fg) = f(x)v(g) + g(x)v(f)\}.$$

**Proposition III.20 ([Per08], Example 1.3).**

$\mathrm{T}_x X$ is a $k$-vector space.

This notion is "functorial", meaning that $f : X \to Y$ a map of algebraic varieties gives a map $\mathrm{T}_x f : \mathrm{T}_x X \to \mathrm{T}_{f(x)} Y$.

**Definition III.63.**

We say that an algebraic variety $X$ is *smooth* if every point $x$ is non-singular, meaning that $\dim(X) = \dim_k(\mathrm{T}_x X)$.

Informally, a smooth algebraic variety does not contain any cusps and does not intersect itself.
We also have a notion of smoothness on morphisms:

**Definition III.64 ([Per08], see Problem VI.1).**

Let $f : X \to Y$ a map between smooth algebraic varieties.
$f$ is said to be *smooth* if $f(X)$ is (Zariski-)dense in $Y$, and if for every $x \in X$, the tangent map $\mathrm{T}_x f$ is surjective.

This definition suffices since our varieties will be assumed to be smooth, but a general definition exists.

### III.4.5 Sheaves of modules and coherence

We are interested to some $\mathcal{O}_X$-modules that generalize the notion of vector bundles, as we cannot define them the same way as differentiable of complex manifold.

**Proposition III.21 ([Per08], Definition 7.1).**

Let $V$ be an affine algebraic variety, set $A = \Gamma(V, \mathcal{O}_V)$.
The functor $M \mapsto \tilde{M}$ from the category of $A$-modules to the one of $\mathcal{O}_V$-modules, defined on every standard open $D(f)$ by $\tilde{M}(D(f)) := M[1/f] = M \otimes_A A[1/f]$, is well defined.

**Definition III.65.**

The objects in the essential image of the above functor are called *quasi-coherent sheaves*. Moreover, if it is isomorphic to a $\mathcal{O}_V$-module $\tilde{M}$ with $M$ finitely generated over $A$, it is called a *coherent sheaf*.

On a general algebraic variety:

**Definition III.66.**

Let $X$ be an algebraic variety and $\mathcal{F}$ be a $\mathcal{O}_X$-module.
$\mathcal{F}$ is said to be quasi-coherent (resp. coherent) if there exists an open affine covering $\{U_i,\ i \in I\}$ of $X$ such that $\mathcal{F}|_{U_i}$ is quasi-coherent (resp. coherent) for each $i$.

**Definition III.67.**

Let $X$ be an algebraic variety.
A *vector bundle* on $X$ is a locally free coherent sheaf.

**Definition III.68.**

Let $X$ be an algebraic variety.
The *cotangent sheaf on* $X$ is the sheaf $\Omega^1_X := \delta^{-1}(I/I^2)$, where $\delta : X \to X \times X$ is the diagonal embedding and

$$I(V) := \{f \in \mathcal{O}_{X\times X}(V) \mid f(V \cap \delta(X)) = \{0\}\}$$

on every open $V$ of $X \times X$.
The sections of $\Omega^1_X$ are called *differential forms*.

**Proposition III.22 ([HTT08], Appendix 1.4).**

The sheaf $\Omega^1_X$, equipped with the natural map $\mathcal{O}_X \to \delta^{-1}\mathcal{O}_{X\times X}$ is an $\mathcal{O}_X$-module.
If moreover $X$ is smooth, on an affine chart, it coincides with the free $\mathcal{O}_X$-module generated by $df$, $f \in \mathcal{O}_X$, satisfying:

1. $d(f+g) = df + dg$,
2. $da = 0$ if $a \in k$,
3. $d(fg) = fdg + gdf$.

## III.5 Some cohomology theory

For this section, we follow [[Har77], Chapter 3], [[Voi02], §4.3] and [HM17].
The cohomology theories are an entire world by themselves, especially for concrete computations (Čech, crystalline, étale, Dwork, simplicial ...). We present only what is superficially needed (the proofs would require more cohomology theories).

### III.5.1 Sheaf cohomology

We will consider sheaf of abelian groups in this subsubsection.
As we notice in III.16, the functor $\Gamma := \Gamma(X, -) : \mathrm{Sh}_X \to \mathscr{A}b$ is not exact, but only left exact, for a topological space $X$. We want to apply tools from the derived functors to this particular functor.

First, one has to check that the category of sheaves of abelian groups satisfy the general assumptions, meaning that they have enough injectives:

**Lemma III.23 ([Har77], Corollary 2.3).**

The category $\mathrm{Sh}_X$ of sheaves of abelian groups has enough injectives.

Therefore, we can define:

**Definition III.69.**

We denote by $\mathrm{H}^n(X, \mathcal{F}) := \mathrm{R}^n\Gamma(\mathcal{F})$ the $n$th cohomology group of the sheaf $\mathcal{F}$.

### III.5.2 de Rham cohomology

The two cohomologies presented will be linked by a theorem of Grothendieck later, see VII.2.2.a.

#### III.5.2.a. Analytic de Rham

In this paragraph, $X$ is a complex manifold of dimension $n$.

**Definition III.70.**

The *de Rham complex* associated to $X$ is:

$$\Omega_X^\bullet : \mathscr{O}_X \xrightarrow{d} \Omega_X^1 \xrightarrow{d} \dots \xrightarrow{d} \Omega_X^n,$$

and denote by $\mathrm{H}^\bullet_{\mathrm{dR}}(X)$ the associted cohomology.

**Theorem III.24 (Poincaré's lemma, [Voi02], Proposition 4.18).**

$\Omega_X^\bullet$ is a right resolution of the constant sheaf $\mathbb{C}_X$.

We denote by $\mathrm{H}^\bullet_{\mathrm{dR}}(X)$ the associated cohomology groups.

**Proposition III.25 ([Voi02], Corollary 4.37).**

We have the following:

$$\mathrm{H}^\bullet(X, \mathbb{C}) \simeq \mathrm{H}^\bullet_{\mathrm{dR}}(X).$$

#### III.5.2.b. Algebraic de Rham

In this paragraph, $X$ is a smooth variety over a field $k \subset \mathbb{C}$, of dimension $n$.
The definition of the de Rham complex still holds (with algebraic differentials), but it does not give an acyclic resolution for the Zariski topology. Therefore, we use an exact resolution of this complex, and use the result III.11.

**Definition III.71.**

The *algebraic de Rham cohomology* of $X$ is defined as the hypercohomology

$$\mathrm{H}^\bullet_{\mathrm{dR}}(X) := \mathbb{H}^i(X, \Omega_X^\bullet),$$

meaning the cohomology of the total complex associated to an exact resolution of $\Omega_X^\bullet$.

We will need to change coefficient of the variety $X$ later on, in the following way (that can be stated in a more general way):

**Lemma III.26 (Base change, [HM17], Lemma 3.1.11).**

There is a natural isomorphism:

$$\mathrm{H}^i_{\mathrm{dR}}(X) \otimes_k \mathbb{C} \simeq \mathrm{H}^i_{\mathrm{dR}}(X_{\mathbb{C}}),$$

where $X_{\mathbb{C}}$ denote the variety $X$ "with coefficients in $\mathbb{C}$"[a].

[a] the formal defintion would require the theory of schemes

### III.5.3 Singular cohomology

This theory originally comes from algebraic topology, with the objectives of detecting "holes" in a topological space. We follow [[Ass22], IX§6] and [[HM17], I§2.2].

**Definition III.72.**

Let $n \geq 0$ be a integer.
The *topological $n$-simplex* is defined as

$$\Delta_n := \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1}, \forall i, t_i \geq 0, \text{ and } \sum_{i=0}^{n} t_i = 1\}.$$

Let $R$ be a commutative ring.

**Definition III.73.**

A *singular $n$-simplex* is a continuous map

$$\Delta_n \to X.$$

The group of singular $n$-chains is the free abelian group:

$$S_n(X) = R[f \text{ singular } n\text{-simplex}]$$

and the group of singular $n$-cochains as

$$S^n(X) = \mathrm{Hom}_R(S_n(X), R).$$

**Definition III.74.**

The $n$-th boundary map is defined as

$$\begin{aligned} d_n : \quad S_n(X) &\longrightarrow S_{n-1}(X) \\ f &\longmapsto \sum_{i=0}^{n} (-1)^i f|_{t_i=0} \end{aligned}.$$

and the induced $n$-th coboundary map

$$d^n : S^n(X) \to S^{n+1}(X).$$

**Lemma III.27 ([HM17], Lemma 2.2.3).**

$(S_n(X), d_n)$ and $(S^n(X), d^n)$ are complexes of abelian groups.

**Definition III.75.**

We define singular homology and cohomology with values in $R$ are defined as

$$\mathrm{H}_i^{\mathrm{sing}}(X, R) := \mathrm{H}_i(S_\bullet(X)), \quad \mathrm{H}^i_{\mathrm{sing}}(X, R) := \mathrm{H}^i(S^\bullet(X)).$$

As the de Rham cohomology, relation exists with sheaf cohomology:

**Theorem III.28 ([Voi02], Theorem 4.47).**

Assume that $X$ is locally contractible.
Then,

$$\mathrm{H}^i_{\mathrm{sing}}(X, R) \simeq \mathrm{H}^i(X, R).$$

To compute the singular cohomology with other coefficients, one can use the *Universal coefficient theorem for homology* (see [[Hat02], Theorem 3A.3]), to deduce:

**Proposition III.29.**

Assume that $R$ has characteristic 0.
Then

$$\mathrm{H}_i^{\mathrm{sing}}(X, R) \simeq \mathrm{H}_i^{\mathrm{sing}}(X, \mathbb{Z}) \otimes_{\mathbb{Z}} R.$$

Therefore, by combining the result on de Rham and singular cohomology (with complex coefficients), we have an isomorphism between both.
It is made explicit by the following:

**Theorem III.30 (de Rham, [Voi02], Remark 4.48).**

Let $X$ be a complex manifold.
For every integer $i \geq 0$, the map

$$\begin{aligned} \mathrm{comp} : \ \mathrm{H}^i_{\mathrm{dR}}(X) &\longrightarrow \mathrm{H}^i_{\mathrm{sing}}(X, \mathbb{C}) \\ \omega &\longmapsto \left[\sum a_j \gamma_j \mapsto \sum a_j \int_{\Delta_j} \gamma_j^* \omega\right] \end{aligned}$$

is well defined (by Stokes' theorem), and is an isomorphism.

# IV The analytic Riemann-Hilbert correspondence

## IV.1 Preliminary: a correspondence in algebraic topology

The goal of this subsection is to motivate the role of the Riemann-Hilbert correspondence in a bigger scheme, modeled on the Galois theory for field extensions, and introduce some notions that will be reused in a particular case later. We follow the presentation of [[Sza12], Chapter 2][1]. No proofs will be given.

### IV.1.1 General considerations on cover theory

**Definition IV.1.**

Let $X$ be a topological space.
A *covering space* over $X$ is a topological space $Y$ and a continuous $p : Y \to X$ such that: for each $x \in X$, there exists an open neighbourhood $V$ of $x$ and a discrete space $S$ such that $\pi^{-1}(V) \simeq S \times V$.

**Definition IV.2.**

Let $p : Y \to X$ and $p' : Y' \to X$ two covering spaces.
A *morphism of covers* is a continuous $\phi : Y \to Y'$ such that $p' \circ \phi = p$.

One way of building such a covering is extracted from certains left-action of groups.

**Definition IV.3.**

Let $G$ be a group acting continuously from the left on a topological space $Y$.
This action is said to be *even* if each point $y \in Y$ has some open neigbourhood $U$ such that for each $g \neq h \in G, gU \cap hU = \emptyset$.

**Proposition IV.1 ([Sza12], Lemma 2.1.7).**

If $G$ acts evenly on $Y$, the projection $p : Y \to G\backslash Y$ is a covering.

What follows is often refers as the Galois theory of covers: we study how the information of the (group of) automorphisms of a covering allows its classification.
We denote by $\mathrm{Aut}(Y/X)$ the group of automorphism of a cover $Y \to X$ (respecting the cover structure).
In order for the theory to works well, we assume that the base space $X$ is locally connected.

**Proposition IV.2 ([Sza12], Proposition 2.2.3).**

Let $p : Y \to X$ be a connected cover.
Then the action of $\mathrm{Aut}(Y/X)$ on $Y$ is even.

Conversely:

[1] the textbook actually explains this big scheme, often called the Galois-Grothendieck correpondence.

**Proposition IV.3.**

Let $G$ be acting evenly on a connected space $Y$.
Then $\mathrm{Aut}(Y/(G\backslash Y)) = G$.

Finally, we can state the so-called *Galois correspondence* for covers.

**Definition IV.4.**

A connected cover $p : Y \to X$ is said to be a *Galois cover* if $\mathrm{Aut}(Y/X)$ acts transitively on each fiber of $p$.

**Theorem IV.4 ([Sza12], Theorem 2.2.10).**

Assume that $X$ is locally connected. Let $: Y \to X$ a Galois cover. We set $G = \mathrm{Aut}(Y/X)$.

1) For each subgroup $H$ of $G$, the projection $p$ induces a natural map $\bar{p}_H : H\backslash Y \to X$ which turns $H\backslash Y$ into a cover of $X$.
2) Let $q : Z \to X$ be a cover fitting in the diagram:

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[dr, "p"'] & Z \arrow[d, "q"] \\ & X \end{tikzcd}$$

then $f$ is a Galois cover and $Z \simeq \mathrm{Aut}(Y/Z)\backslash Y$.

3) The maps $H \mapsto H\backslash Y$ and $Z \mapsto \mathrm{Aut}(Y/Z)$ induces a bijection between subgroups of $G$ and intermediate cover of $p$ as above,
4) The intermediate cover $q : Z \to X$ is Galois if, and only if, $H$ is a normal subgroup of $G$, in which case $\mathrm{Aut}(Z/X) \simeq G/H$.

### IV.1.2 The monodromy action

The monodromy action if often introduced in the context of differential equations (and it will the case in our study later on). However, it fits into the general theory of covers.
It starts with a "lifting" lemma:

**Lemma IV.5 ([Sza12], Lemma 2.3.2).**

Let $p : Y \to X$ be a cover, $y \in Y$ and set $x = p(y)$.

1) Let $f : [0, 1] \to X$ be a path with base point $x$: $f(0) = x$. There exists a unique path $\tilde{f} : [0, 1] \to Y$ such that $\tilde{f}(0) = y$ and $p \circ \tilde{f} = f$,
2) Let $g : [0, 1] \to X$ be another path with base point $x$, homotopic to $f$. Then $\tilde{f}(1) = \tilde{g}(1)$ (for the unique $\tilde{g}$ as define above).

Therefore, we construct an action of the fundamental group $\pi_1(X, x)$ on the fiber $p^{-1}(x)$:

**Definition IV.5 ([Sza12], Construction 2.3.3).**

The *monodromy action* of $\pi_1(X,x)$ on $p^{-1}(x)$ as follows: if $\alpha = [f] \in \pi_1(X,x)$ and $y \in p^{-1}(x)$, then $\alpha.y := \tilde{f}(1)$.

**Remark IV.1.**

It is either a left or a right action, depending on the convention one adopt for the multiplication for paths. Let us assume that this is a left one.

This construction induces a functor:

**Definition IV.6.**

Let $x \in X$.
We define a functor

$$\begin{array}{rccl} \mathrm{Fib}_x : & \{p : Y \to X \text{ cover}\} & \longrightarrow & \{\text{set with a left-}\pi_1(X,x) \text{ action}\} \\ & p : Y \to X & \longmapsto & p^{-1}(x) \end{array} .$$

The morphisms are respected by the definition of morphisms of covers.

**Theorem IV.6 ([Sza12], Theorem 2.3.4).**

Assume that $X$ is connected and locally simply connected, let $x \in X$.
Then the functor $\mathrm{Fib}_x$ induces an equivalence of categories. Moreover, connected covers correspond to sets with a transitive action, and Galois cover to coset spaces of normal subgroups.

The proof comes with the construction of "the" *universal cover*:

**Theorem IV.7 ([Sza12], Theorem 2.3.5).**

Assume that $X$ is connected and locally simply connected, let $x \in X$.
Then the functor $\mathrm{Fib}_x$ is representable by a cover $\tilde{X}_x$, called the *universal cover above $x$*.

In the other hand, in line with III.16, we have:

**Proposition IV.8 ([Sza12], Theorem 2.5.9).**

The functor $Y \mapsto \mathcal{F}_Y$ from the category of covers over $X$ to the one of locally constant sheaves, obtained by taking the sheaf of sections over $Y$, induces an equivalence of categories.

Mixing both results, we can restrict the functor $\mathrm{Fib}_x$ to the following:

**Theorem IV.9 ([Sza12], Corollary 2.6.2).**

Assume that $X$ is connected and locally simply connected, let $x \in X$.
The functor $\mathrm{Fib}_x$ induces an equivalence between the category of complex local systems, *ie* locally constant sheaves of finite dimensional complex vector spaces, and the one of finite dimensional left representations of $\pi_1(X,x)$.

We will denote $\mathrm{LocSys}(X)$ the category of local systems on $X$.
Such a representation is called the *monodromy representation* of the local system.

It coincides with the monodromy in the sense of differential equations. Indeed, let us consider the following differential equation on a domain $U \in \mathbb{C}$:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = 0,$$

with the $a_i$'s holomorphic on $U$. By a fundamental theorem of Cauchy, the solutions on an open neighourhood of each point is a finite dimensional vector space. Thus, the sheaf of solutions is a local system. By the previous theorem, it is associated to a monodromy representation of the fundamental group $\pi_1(U)$.

## IV.2 Holomorphic version on a complex manifold

Some ideas of this paragraph are taken from [[Voi02], §9.2], [[Sza12], §2.7] and [[Poo25], §7.7].

**Theorem IV.10 (Riemann-Hilbert correspondence, complex manifold case).**

Let $X$ be a complex manifold.
The functor

$$\begin{array}{rcl} F: & \mathrm{LocFree}^{\nabla}(X) & \longrightarrow \mathrm{LocSys}(X) \\ & (\mathcal{E}, \nabla) & \longmapsto \mathcal{E}^{\nabla} := \mathrm{Ker}(\nabla) \end{array}$$

induces an equivalence of categories.
Its inverse is given by

$$\begin{array}{rcl} G: & \mathrm{LocSys}(X) & \longrightarrow \mathrm{LocFree}^{\nabla}(X) \\ & \mathcal{A} & \longmapsto (\mathscr{O}_X \otimes_{\mathbb{C}} \mathcal{A}, d \otimes 1) \end{array}$$

Proof

Step 1: $G$ is well defined, that is: if $\mathcal{A} \in \mathrm{LocSys}(X)$, then $(\mathcal{E}, \nabla) := (\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X, 1 \otimes d) \in \mathrm{LocFree}^{\nabla}(X)$.

First remark that $d \otimes 1$ is well defined by extension of $\mathscr{O}_X \to \Omega^1_X$ with the tensor product. We also use the identification

$$(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X) \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \mathcal{A} \otimes_{\mathbb{C}} (\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X) \simeq \mathcal{A} \otimes_{\mathbb{C}} \Omega^1_X$$

by associativity of the tensor product.
For a small open set such that $\mathcal{A}|_U \simeq \mathbb{C}^r$, we have $(\mathcal{A} \otimes \mathscr{O}_X)|_U \simeq \mathcal{A}|_U \otimes \mathscr{O}_U \simeq \mathbb{C}^r \otimes \mathscr{O}_U \simeq \mathscr{O}_U^{\oplus r}$, so it is locally free.
Finally, we have:

$$\nabla^{(1)} \circ \nabla(a \otimes f) = \nabla^{(1)}((a \otimes 1) \otimes df) = \nabla(a \otimes 1) \wedge df + a \otimes d(df) = (a \otimes d1) \wedge df = 0.$$

We again used that the image of $a \otimes f$ in $\mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ is identified with $(a \otimes 1) \otimes df$.
We also could have use the form of the connections on $\mathscr{O}_U^r$, see III.17.
Step 2: $G$ is a functor.
If $\phi : A \to B$ is a morphism in $\mathrm{LocSys}(X)$, we have an induced $G(\phi) = \phi \otimes 1 : A \otimes_{\mathbb{C}} \mathscr{O}_X \to B \otimes_{\mathbb{C}} \mathscr{O}_X$ that makes the following commute:

$$\begin{array}{ccc} A \otimes_{\mathbb{C}} \mathscr{O}_X & \xrightarrow{\phi\otimes 1} & A \otimes_{\mathbb{C}} \mathscr{O}_X \\ {\scriptstyle 1\otimes d}\downarrow & & \downarrow{\scriptstyle 1\otimes d} \\ A \otimes_{\mathbb{C}} \Omega_X^1 & \xrightarrow{\phi\otimes 1} & B \otimes_{\mathbb{C}} \Omega_X^1 \end{array}$$

Step 3: $F$ is well defined, that is: if $(\mathcal{E}, \nabla) \in \mathrm{LocFree}(X)$, then $\mathcal{E}^\nabla \in \mathrm{LocSys}(X)$.
This is the most difficult part. Let us sketch out the proof.
The result is equivalent to saying that *locally*, the solutions of our system of differential equations given by $\nabla$ is a complex vector space, with dimension equal to the rank of the system. One can then think of the Cauchy-Lipschitz (or Picard-Lindelöf) theorem. It indeed works when $X$ is a curve because we have locally an ordinary differential equation. If $n > 1$, this theorem cannot be applied.
There is an alternative for this type of system: the Frobenius theorem. One corollary of this theorem is the following lemma, that implies immedialety the result.

**Lemma IV.11 ([Voi02], Lemma 9.12).**

Let $x \in X$.
There exists an open set $U$ containing $x$ such that the evaluation $\mathrm{ev}_x : \mathrm{Ker}(\nabla|_U) \to \pi^{-1}(x)$ is an isomorphism (of sheaves of complex vector spaces).

Step 4: $F$ is a functor.
If $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^\nabla(X)$, then it induces a morphism of sheaves $\phi : \mathrm{Ker}(\nabla_1) \to \mathcal{E}_2$, and its image is a subset of $\mathrm{Ker}(\nabla_2)$ by the relation $\nabla_2 \circ \phi = (\phi \otimes 1) \circ \nabla_1$.
Step 5: $GF \simeq 1_{\mathrm{LocFree}^\nabla(X)}$.
Let $(\mathcal{E}, \nabla) \in \mathrm{LocFree}^\nabla(X)$. Set

$$\alpha_{(\mathcal{E},\nabla)} : \begin{array}{ll} \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X & \longrightarrow \mathcal{E} \\ s \otimes f & \longmapsto fs \end{array}.$$

It is clearly well defined and for $s \otimes f \in \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X$, we have, using the Leibniz rule:

$$\nabla(\alpha_{(\mathcal{E},\nabla)}(s \otimes f)) = \nabla(fs) = s \otimes df = \alpha(s \otimes 1) \otimes df = (\alpha \otimes 1)((1 \otimes d)(s \otimes f)),$$

so $\alpha_{(\mathcal{E},\nabla)}$ define a morphism in the category $\mathrm{LocFree}^\nabla(X)$.
Better, it defines a natural transformation. Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^\nabla(X)$. Then,

$$\phi(\alpha_{(\mathcal{E}_1,\nabla_1)}(f \otimes s)) = \phi(fs)$$

and

$$\alpha_{(\mathcal{E}_2,\nabla_2)}(GF(\phi)) = \alpha_{(\mathcal{E}_2,\nabla_2)}(F(\phi)(s) \otimes f) = f\phi(s).$$

This is equal because $\phi$ is a morphism of $\mathscr{O}_X$-modules.
To show that this is an isomorphism, let $U$ be a small open neighborhood of $x$, such that $\mathrm{Ker}(\nabla|_U) \simeq \mathbb{C}^r$. Let $(s_1, \ldots, s_r)$ a basis of $\mathrm{Ker}(\nabla|_U)$. Then, $(s_1 \otimes 1, \ldots, s_r \otimes 1)$ is a basis of $\mathrm{Ker}(\nabla|_U) \otimes_{\mathbb{C}} \mathscr{O}_U$.
Moreover by taking $U$ small enough, $(s_1, \ldots, s_r)$ is also a $\mathscr{O}_U$-basis of $\mathcal{E}(U) \simeq \mathscr{O}_U^{\oplus r}$. Indeed,

the determinant of the matrix (in any basis of $\mathcal{E}_U$) with $j$-th column $(s_j^{(i)})^T$ is not zero at $x$, so does not vanish in a small neighborhood of $x$.
Therefore, the matrix of $\alpha_{(\mathcal{E},\nabla)}$ in the basis $(s_1, \dots, s_r)$ (on both sides) is the identity, and is invertible.
Thus, $\alpha_{(\mathcal{E},\nabla)}$ is invertible on each stalk, and it is an isomorphism of sheaves.
Step 6: $FG \simeq 1_{\mathrm{LocSys}(X)}$.
We clearly have for a local system $\mathcal{A}$ on $X$, $(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X)^{1\otimes d} \simeq \mathcal{A} \otimes_{\mathbb{C}} \mathbb{C} \simeq \mathcal{A}$, and it behaves well with morphisms, such that it defines an isomorphism of functors.

$\square$

## IV.3 Meromorphic version on a complex manifold

For a Riemann surface (meaning, $\dim(X) = 1$), the previous result gives us that any finite dimensional representation of $\pi_1(X, x_*)$ can be obtained as the monodromy reprensentation of some linear system of holomorphic differential equations.
At the begenning of the twentieth century, Hilbert stated, in one of his "23 problems of the century" (precisely, the twenty-first) a more precise question: is it possible to obtain such a finite dimensional representation of the fundamental group as the monodromy group of a differential equation with "moderated-growth solutions" around poles, for say $X$ equal to $\hat{\mathbb{C}} \setminus S$, with $S$ a finite set of points.
We develop this point of view in this section.

### IV.3.1 In dimension 1

#### IV.3.1.a. Elements of Fuchs theory

We basically follow Haefliger's lecture in [Bor+87], and the theory presented is usually refered to as the *Fuchs theory.* As in III.2, we add some more examples and proofs, as I gave a talk on the subject during the internship.

Let $X = \hat{\mathbb{C}} \setminus S$, with $S = \{s_1, \dots, s_k\}$.
We focus out study around each point of $S$. Without loss of generality, the case around $z = 0$ is sufficient. We will denote $D_\varepsilon$ the disk centered at 0 of radius $\varepsilon$ and $D_\varepsilon^*$ the associated punctured disk at $z = 0$.

If we wanted to translate the definition of a connection on $D_\varepsilon$, we would say that such a connection is a locally free sheaf of modules $\mathcal{E}$ over the sheaf of rings of meromorphic function defined on $D_\varepsilon$ with only a pole at 0, equipped with a map $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_{D_\varepsilon}} \Omega^1_{D_\varepsilon}$ with linearity and Leibniz properties.
We can assume that $\mathcal{E}$ is free. Moreover, it is more convenient to work with the following:

**Definition IV.7.**

We denote by $K$ the field of germs of meromorphic functions around 0.

Having localised our study, we then will not work with sheaf language in this paragraph. This shift of persective leads us to the following definition.

**Definition IV.8.**

A (germ of) *meromorphic connection* at 0 is a finite dimensional vector space $M$ over $K$ equipped with a $\mathbb{C}$-linear operator $\nabla : M \to M$ satisfying the Leibniz rule:

$$\forall (h,u) \in K \times M : \nabla(hu) = \frac{dh}{dz} u + h\nabla u.$$

With this definition, a morphism of connection is of linear morphism commuting with connections.

**Remark IV.2.**

The ring $D[z^{-1}]$ of germs of differential operators with meromorphic coefficients acts on a meromorphic connection $(M, \nabla)$ by $Pu = \sum_{j=1}^{N} a_j \nabla^j u$ if $P = \sum_{j=1}^{N} a_j \frac{d^j}{dz^j}$ and $u \in M$.

**Remark IV.3.**

As explained in III.23, given a basis of $M$, a connection map on $M$ is equivalent to a system of differential equations: a meromorphic connection is given by $\nabla = d - A dz$, where $A$ is a matrix with coefficients in $K$, and vice versa.
A morphism between systems of equation given by matrices $A$ and $B$ respectively is then a matrix $M : D_\varepsilon^* \to \mathrm{M}_n(K)$ such that $\frac{dM}{dz} = BM - MA$, and we say that two systems of differential equations are *equivalent* if $M(z)$ is invertible for any $z \in D_\varepsilon$ (or equivalently, the associate connections are isomorphic).
In the rest of the paragraph, we will juggle between these two definitions.

Let us compute the monodromy associated to a differential system $y' = Ay$ on $D_\varepsilon$.
First, recall that the universal cover of $D_\varepsilon^*$ given by the exponential is $\{t \in \mathbb{C}, |e^{2i\pi t}| < \varepsilon\}$. Multivalued functions are thus defined on this universal cover. The function $t$ and $\tilde{\cdot}$ will belong to the universal cover while $z$ will belong to $D_\varepsilon$.
Let $\tilde{S}(t) = (\tilde{U}_1, \ldots, \tilde{U}_n)$ a fundamental system of multivalued solutions, meaning that form a basis of the vector space of solutions. The matrix $A$ is 1-periodic on the universal cover, so $t \mapsto \tilde{S}(t+1)$ is also a solution. Thus, there exists a constant invertible matrix $C$ such that $\tilde{S}(t+1) = \tilde{S}(t)C$.

**Definition IV.9.**

The matrix $C$ defined above is called the *monodromy matrix* of the system.

To get back in the punctured disk, let $\Gamma$ a matrix such that $C = \exp(2\pi i \Gamma)$. Then, $\tilde{S}(t)\exp(-2\pi i t \Gamma)$ is 1-periodic on the universal cover, and one can well define $\Sigma(z) = \tilde{S}(\frac{1}{2\pi i}\log(z))e^{\log(z)\Gamma}$. Finally, we set $S(z) = \Sigma(z)e^{\log(z)\Gamma}$.

One cannot expect to have a correspondence between the category of meromorphic connections and the one of representations of $\pi_1(D_\varepsilon^*)$, as the following example shows:

**Example IV.4.**

Define the two connections $\nabla_1 = d$ and $\nabla_2 = d - \frac{dz}{z^2}$. They are not equivalent since the matrix between the two systems is $M(z) = Ce^{1/z}$, and it is not defined at 0 (thus not invertible), but they both have the trivial monodromy (the solutions of $\nabla_2$ are of the form $Ce^{-1/z}$).

This phenomenon can be intuited as resulting of a *bad* behavior of the solutions at $z = 0$, meaning that they have there sort of an essential singularity.
Thus, we introduce a *nice* class of functions.

**Definition IV.10.**

Let $f$ be a multivalued function around 0.

$f$ has *moderate growth* if, for all sector $S^{\eta}_{(a,b)} := \{z \in D^*_\eta \mid \arg(z) \in (a,b)\}$ over which $f$ is defined, there exist $k \in \mathbb{N}$ and $C > 0$ such that:

$$\forall z \in S^{\eta}_{(a,b)},\ |f(z)| \leq C|z^{-k}|.$$

For a $n$-uplet of functions, the definition is the same, with a norm on the left hand side (any one of them, they are equivalent).

**Example IV.5.**

1) If $f$ is a single-valued function, $f$ has moderate growth if, and only if $f$ is meromorphic on $D_\varepsilon$ (for $\varepsilon$ small enough).
2) $z^\alpha$ for $\alpha \in \mathbb{C}$, $\log(z)^m$ for $m \in \mathbb{N}$, have moderate growth.

The following result if very powerful: it gives you information on the behavior of the solutions of a differential equation just by checking a condition on the coefficients of the differential operator with a very simple computation.

**Theorem IV.12 (Fuchs, [Bor+87], Theorem 1.1.1).**

Let $P = \sum_{k=0}^{n} a_k(z) \frac{d^{n-k}}{dz^{n-k}}$ a differential operator on $D_\varepsilon$ with the $a_k$'s meromorphic.
The following statements are equivalent:

i) the solutions of $Pu = 0$ have moderate growth,
ii) for every $1 \leq i \leq n$, $\frac{a_i}{a_0}$ has a pole of order at most $i$ at 0.

These systems are those we are interested in for an expected correspondence.

**Definition IV.11.**

A differential operator is said to be *regular* if it satifies the conditions of the previous theorem.

**Example IV.6.**

As expected, the example given in IV.4 is not regular.

We can extend this notion for a system of differential equations :

**Definition IV.12.**

Let $y' = A(z)y$ be a system of linear differential equations.
It is said to be a *regular system* if its solutions have moderate growth.

**Theorem IV.13 ([Bor+87], Theorem 1.3.1).**

Let $(E) : y' = Ay$ a differential system on $D_\varepsilon$. The followings are equivalent:

i) $(E)$ is equivalent to a regular system,
ii) $(E)$ is equivalent to a system $y' = \frac{B(z)}{z}y$ for some matrix $B$ holomorphic on $D_\varepsilon$,
iii) $(E)$ is equivalent to a system $y' = \frac{\Gamma}{z}y$ for some constant matrix $\Gamma$.

With this theorem, the equivalent definition in terms of connections is the following:

**Definition IV.13.**

Let $(M, \nabla)$ be a meromorphic connection.
It is said to be an *regular connection* if there exists a basis $(e_1, \ldots, e_n)$ of $M$ over $K$, holomorphic functions $(b_{i,j}(z))$ and an $\varepsilon > 0$ such that:

$$\forall i \in [\![1, n]\!], \forall z \in D_\varepsilon,\ z\nabla e_i = -\sum_{j=1}^{n} b_{i,j}(z) e_j.$$

The following theorem is what we were aiming for.

**Theorem IV.14 ([Bor+87], Corollary 1.3.2).**

Two regular systems on $D_\varepsilon$ are equivalent if, and only if, their monodromy is conjugated.

Stated like that, it is not as precise as we would have wanted for an equivalence of categories like in IV.2. The next paragraph will make it more accurate, based on the theorem above.

First, a result on the good behavior of algebraic construction (see III.18) with regularity:

**Proposition IV.15 ([HTT08], Proposition 5.1.11).**

Let $M$ and $N$ be regular meromorphic connections.
Then $\mathrm{Hom}_K(M, N)$ and $M \otimes_K N$ are regular.

PROOF

We do the full computation on $\mathrm{Hom}_K(M, N)$, equipped with $\nabla : \phi \mapsto \nabla_N \circ \phi - \phi \circ \nabla_M$. The second one is similar.
We set the following notations: $\dim(M) = m, \dim(N) = n$, equipped respectively with basis $(e_1, \ldots, e_m)$ and $(e'_1, \ldots, e'_n)$, $\phi = \sum_{i,j} \phi_{i,j}\varepsilon_{i,j}$, with $\varepsilon_{i,j}(e_k) = \delta_{i,k}e'_j$ a basis of $\mathrm{Hom}_K(M, N)$, and $\nabla_M = \sum_{i,j} \frac{1}{z} b_{i,j}\varepsilon^M_{i,j}, \nabla_N = \sum_{i,j} \frac{1}{z} c_{i,j}\varepsilon^N_{i,j}$ (the $\varepsilon^M_{i,j}$ and $\varepsilon^N_{i,j}$ are defined analoguously, and the index does not range on the same interval, but the reader can write down the details if not convinced).
Therefore:

$$\begin{aligned}
\nabla(\phi) &= \left(\sum_{i,j} \frac{1}{z} c_{i,j}\varepsilon^N_{i,j} \circ \sum_{k,l} \phi_{k,l}\varepsilon_{k,l}\right) - \left(\sum_{i,j} \phi_{i,j}\varepsilon_{i,j} \circ \sum_{k,l} \frac{1}{z} b_{k,l}\varepsilon^M_{k,l}\right) \\
&= \sum_{i,j,k,l} \frac{1}{z}\left(c_{i,j}\phi_{k,l}\varepsilon^N_{i,j} \circ \varepsilon_{k,l}\right) - \sum_{i,j,k,l} \frac{1}{z}\left(\phi_{i,j}b_{k,l}\varepsilon_{i,j} \circ \varepsilon^M_{k,l}\right) \\
&= \sum_{i,j,k,l} \frac{1}{z} c_{i,j}\phi_{k,l}\delta_{i,l}\varepsilon_{k,j} - \sum_{i,j,k,l} \frac{1}{z}\phi_{i,j}b_{k,l}\delta_{i,l}\varepsilon_{k,j} \\
&= \sum_{j,k} \frac{1}{z}\left(\sum_i c_{i,j}\phi_{k,i} - \phi_{i,j}b_{k,i}\right)\varepsilon_{k,j}.
\end{aligned}$$

One can see that the connection obtained is of the desired form.

□

#### IV.3.1.b. The local lifting

This paragraph is partially inspired by [[Sza12], §2.7], and [[HTT08], Lemma 5.2.19] for the lifting of morphisms.
The precise Riemann-Hilbert correspondence will be stated next paragraph, based on what we already did in the holomorphic case. In this one, we try to construct a unique (regular) meromorphic connection on $D_\varepsilon$ given a holomorphic connection on $D_\varepsilon^*$.
We return to the language of sheaves. Indeed, according to a general fact on non-compact Riemann surfaces, a locally free $\mathscr{O}_{D_\varepsilon^*}$-module on $D_\varepsilon$ is actually free (for example, see [[For81], Theorem 30.4]).
We will denote by $\mathscr{O}_{D_\varepsilon}(*0)$ the sheaf of meromorphic functions on $D_\varepsilon$ with a pole at $z = 0$.

**Proposition IV.16.**

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on $D_\varepsilon^*$.
There exists a regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on $D_\varepsilon$ which agrees with $(\mathcal{E}, \nabla)$ on $D_\varepsilon$.
This connection is unique up to isomorphisms (of meromorphic connections).

Proof

We can assume $\mathcal{E} = \mathscr{O}_{D_\varepsilon}^r$ and $\nabla = d - Adz$. Set $S(z)$ a fundamental system of solutions of multivalued functions on $D_\varepsilon^*$. Let $C$ the monodromy matrix associated to $\nabla$, $S(z)$ a fundamental system of solutions, and let $B$ a constant matrix such that $C = \exp(2i\pi B)$. We consider the regular connection $\bar{\nabla} = d - \frac{B}{z}dz$. A fundamental system of solutions is given by $z^B = e^{B\log(z)}$.
Let $U(z) = S(z)z^{-B}$. Then

$$U(ze^{2i\pi}) = S(ze^{2i\pi})e^{-B\log(ze^{2i\pi})} = S(z)Ce^{-2i\pi B}z^{-B} = U(z).$$

Here, by an abuse of notation, $ze^{2i\pi}$ denotes the composition by the simple loop going around 0 counter-clockwise.
Therefore, $U(z)$ is single-valued, and then holomorphic on $D_\varepsilon^*$. Its inverse is too, so it defines a isomorphism between $(\mathcal{E}, \nabla)$ and $(\mathscr{O}_{D_\varepsilon}(*0)^r, \bar{\nabla})$.
Uniqueness comes from IV.14.

□

To make a functorial statement, we need additionally an extension of morphisms, we will directly do it for the global framework.

#### IV.3.1.c. Global study via patching

We will state the "meromorphic" Riemann-Hilbert on $\mathbb{P}^1_{\mathbb{C}}$ minus a finite set of points. We firstly introduce some global notation compared with what we did the last two paragraphs, and then state and prove the result.
Some ideas comes from [Sza12].

In this paragraph, we set $X = \mathbb{P}^1_{\mathbb{C}}$, $D = \{s_1, \ldots, s_k\} \subset X$ and $Y = X \setminus D$.
We denote by $\mathscr{O}_X(*D)$ the sheaf of meromorphic functions on $X$ with poles along $D$.

**Definition IV.14.**

Let $\bar{\mathcal{E}}$ a locally free $\mathscr{O}_X(*D)$-module.
A *meromorphic connection* on $\bar{\mathcal{E}}$ is a $\mathbb{C}$-linear map $\bar{\nabla} : \bar{\mathcal{E}} \to \bar{\mathcal{E}} \otimes_{\mathscr{O}_X(*D)} \Omega^1_X$ satisfying the Leibniz

rule.

**Definition IV.15.**

We define the category $\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D)$ with

- locally free $\mathscr{O}_X(*D)$-modules equipped with a regular meromorphic (flat) connection as objects,
- morphisms of $\mathscr{O}_X(*D)$-modules commuting with the connections as morphisms.

**Theorem IV.17.**

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on $Y$.
There exists a regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on $X$ which agrees with $(\mathcal{E}, \nabla)$ on $Y$.
This connection is unique up to isomorphisms (of meromorphic connections).

Proof

Let $D_i$ small disks around each $s_i$ that do not meet. On each $D_i$, let $(\bar{\mathcal{E}}_i, \bar{\nabla}_i)$ the unique (up to isomorphisms) lifting of $(\mathcal{E}_i, \nabla_i) := (\mathcal{E}, \nabla)|_{D_i}$ given by IV.16. On an open set $U$ containing the complementary of $\bigcup_{i=1}^{k} D_i$ and which does not meet $S$, we set $\bar{\mathcal{E}} = \mathcal{E}$.
Then, we can glue back the sheaves on $X = \bigcup_{i=1}^{k} D_i \cup U$ by III.15 (the isomorphisms $\theta_{i,j}$ are just given by the ones coming from the construction of the $\bar{\mathcal{E}}_i$, and the triple intersection condition is empty), which gives us the sheaf $\bar{\mathcal{E}}$ on $X$ that agrees with $\mathcal{E}$ on $Y$.
The uniqueness is given by the one on each $D_i$ and the one on the patching result.

$\square$

Now, we deal with morphisms:

**Proposition IV.18 ([HTT08], Lemma 5.2.19).**

Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism between holomorphic connections on $Y$.
Then, there exists a morphism of meromorphic connections $\bar{\phi} : (\bar{\mathcal{E}}_1, \bar{\nabla}_1) \to (\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ on $X$ that restricts to $\phi$.

We need two intermediate results.

**Lemma IV.19.**

The restriction map given by IV.17 induces a bijection:

$$\Gamma(X, \mathrm{Ker}(\bar{\nabla})) \xrightarrow{\sim} \Gamma(Y, \mathrm{Ker}(\nabla)).$$

Proof

The injectivity is clear. For the surjectivity, let $s \in \Gamma(Y, \mathrm{Ker}(\nabla))$. The connection $\bar{\nabla}$ is regular, so $s$, a solution of the differential equation induced is single-valued as being defined on the whole of $Y$, and therefore has moderate growth as noticed in IV.5 - meaning that it can be extended to a meromorphic section on $D$. This concludes.

$\square$

This one follows the definition of the connection on Hom.

**Lemma IV.20.**

Let $(\bar{\mathcal{E}}_1, \bar{\nabla}_1)$ and $(\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ two connections on $D_\varepsilon$.
Then,

$$\Gamma(X, \mathcal{H}om_{\mathscr{O}_X(*D)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)^{\bar{\nabla}}) \simeq \mathrm{Hom}_{\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X,D)}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)),$$

where the $\bar{\nabla}$ on the left-hand side is the one associated to the sheaf $\mathcal{H}om$ (see III.18).

PROOF (IV.18)

Set $\mathcal{E} = \mathcal{H}om_{\mathscr{O}_Y}(\mathcal{E}_1, \mathcal{E}_2)$ and $\bar{\mathcal{E}} = \mathcal{H}om_{\mathscr{O}_X(*S)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)$ with the associate $\nabla$ and $\bar{\nabla}$. By IV.15, $\bar{\mathcal{E}}$ is regular, so we can apply the first lemma :

$$\Gamma(X, \mathcal{H}om_{\mathscr{O}_X(*D)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)^{\nabla}) \xrightarrow{\sim} \Gamma(X, \mathcal{H}om_{\mathscr{O}_Y}(\mathcal{E}_1, \mathcal{E}_2)^{\nabla}).$$

The second one then gives a bijection:

$$\mathrm{Hom}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)) \simeq \mathrm{Hom}((\mathcal{E}_1, \nabla_1), (\mathcal{E}_2, \nabla_2)),$$

the surjectivity of which ensures the result.

□

**Theorem IV.21.**

The restriction functor

$$\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D) \to \mathrm{LocFree}^{\nabla}(Y)$$

induces an equivalence of categories.

PROOF

The pseudo-inverse functor is given by the construction above, and all the other verifications are clear.

□

With the work done the previous subsection, we have the correspondence as wished:

**Corollary IV.22 (Riemann-Hilbert correspondence, analytic meromorphic case, dimension 1).**

There is an equivalence of categories:

$$\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D) \cong \mathrm{LocSys}(Y).$$

### IV.3.2 Sketch of a proof in higher dimension

The classical reference of this subsubsection is [Del70], and the presentation by Malgrange in [[Bor+87], Part IV] and restructured in [[HTT08], §5.2].

For the general case, we work with a pair $(X, D)$, where $X$ is a connected complex manifold of dimension $n$ and $D$ a divisor of $X$. We set $Y = X \setminus D$.

The main goal is to build a unique regular meromorphic extension of connection on $X$ given a holomorphic one on $Y$ (see [[HTT08], Theorem 5.2.17]), called the *Deligne's canonical extension*. Notice that the central point in the work of Deligne is to redefine the regularity condition for higher dimension. Two approaches are possible: reduce to the 1-dimensional case by verifying the regularity on the pullback of the connection by some $D(0,1) \hookrightarrow X$, or to control the growth of the solutions on the associated PDE. The two are equivalent. See [[Del70], II§4, Theorem 4.1] for details.
The existence is somehow similar to what we have already done in the one-dimensional case, meaning that the can patch local lifting, and actually restrict to the case where $Y = (D^*)^r \times D^{n-r}$, where $D = D(0, \varepsilon)$, and we translate the problem in terms of monodromy representation to construct a suitable connection. This simplification for the form of $Y$ however suits for $D$ a *normal crossing* divisor: a divisor $D$ is said to be of normal crossing if for every point $p \in D$, there exists local coordinates $(x_1, \dots, x_n)$ of a neighbourhood $U$ of $p$ in $X$ such that $D \cap U = \{x_1 \dots x_m = 0\}$.
Fortunatly, one can always refer back to this case by the Hironaka's resolution of singularities (see [Hir64]): there exists a map $f : X' \to X$, proper and surjective, such that $D' := f^{-1}D$ is a normal crossing divisor, and the restriction of $f : X' \setminus D' \to X \setminus D$ is an isomorphism. Therefore, the pullback of a connection $M$ on $X$, $g^*M$, can be extended to a regular meromorphic connection on $X'$, and can give back a regular connection on $X$ by applying some functors and some additional work (for the details, see [HTT08], Theorem 5.2.20).

The uniqueness requires more constraints, namely the log-poles of the connection around $D$ (which is a stronger condition than regularity), and the real-part of the eigenvalues associated to the connection being in a lenght-1 interval. But these constraints are actually useful to glue back local lifting, and have some control over the morphisms between connections.

The result we finally get is the following (we do not state the original one in [Del70] because it's already mixed with the algebraic case):

**Theorem IV.23 (Riemann-Hilbert correspondence, analytic meromorphic case, [HTT08], Corollary 5.2.21).**

Let $X$ be a complex manifold and $D$ a divisor on $X$. Set $Y = X \setminus D$.
Then we have an equivalence:

$$\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D) \cong \mathrm{LocSys}(Y).$$

# V The algebraic Riemann-Hilbert correspondence

This section is rather short. The main reason is that it is based on the previous study on analytic space : thanks to the *GAGA principle*, briefly explained in the first subsubsection, one can, provided a correct translation of the analytic notion of regularity into an algebraic one.
Again, the original paper is the one by Deligne [Del70], and we take some clarification in [[HTT08], 5.3] and [[Bor+87], IV.7].

## V.1 Elements of GAGA

"GAGA" usually refers to an article by Serre [Ser56], "Géométrie algébrique et géométrie analytique". The starting point is the following: given a projective algebraic variety over $\mathbb{C}$, one could consider its algebraic structure (with regular functions) or its analytic structure (with holomorphic functions). Both of these studies gave in general similar results. So the following question arose: is there a systematic way to translate results in the algebraic setup into analytic ones, and vice versa?
This paradigm is still active in the current research. Here, we only state the results that we need. For more, the reader could take a look at [Ser56] or [[GR71], Exposé XII] (in the language of schemes).

**Proposition V.1.**

There exists a functor $(-)^{\mathrm{an}}$ from the category of complex algebraic varieties to the category of analytic spaces, and a functor from the category of algebraic sheaves on $X$ and analytic sheaves on $X^{\mathrm{an}}$, via the map $\mathcal{F}^{\mathrm{an}} = \mathcal{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X^{\mathrm{an}}$.
Moreover, for every $(X, \mathcal{O}_X)$ complex algebraic variety, there exists a morphism of ringed space $(X^{\mathrm{an}}, \mathscr{O}_X^{\mathrm{an}}) \to (X, \mathcal{O}_X)$ (see III.38 for the definition).

From a complex algebraic variety $(X, \mathscr{O}_X)$ (equipped with the Zariski topology), the analytic space $X^{\mathrm{an}}$ consists of the same set of points, but equipped with the Euclidian topology, and the sheaf of regular functions seen as holomorphic functions $\mathscr{O}_X^{\mathrm{an}}$. From a morphism $f : X \to Y$ of algebraic variety, we associate $f^{\mathrm{an}} : X^{\mathrm{an}} \to Y^{\mathrm{an}}$.

**Theorem V.2 ([Ser56], Theorem 2,3).**

Let $X$ be a projective algebraic variety.
The analytification on coherent sheaves is an equivalence of categories.

**Theorem V.3 ([Ser56], Theorem 1).**

For every coherent sheaf $\mathcal{F}$ and every $i \geq 0$, we have:

$$\mathrm{H}^i(X, \mathcal{F}) \simeq \mathrm{H}^i(X^{\mathrm{an}}, \mathcal{F}^{\mathrm{an}}).$$

We only stated the Riemann-Hilbert correspondence for complex manifold so we only work with the following restriction:

**Proposition V.4 ([Har77], Appendix B §1).**

The analytification functor on algebraic varieties restricts to a functor between smooth algebraic varieties and complex manifolds.

## V.2 Meromorphic Riemann-Hilbert on a smooth algebraic variety

In this subsection, $X$ will denote a complex smooth algebraic variety.

Let us establish the framework. Consider $j : X \hookrightarrow V$ an open embedding of $X$ into a smooth algebraic variety $V$, such that $D = V \setminus X$ is a divisor (the definition is similar to the one on a complex manifold). We study integrable connections $(\mathcal{E}, \nabla)$ on $X$, meaning a vector bundle on $X$ as defined in III.4, with a $\mathbb{C}$-linear map $\nabla : \mathcal{E} \to \mathcal{E} \otimes \Omega^1_X$ satisfying the Leibniz rule. Meromorphic connections are defined the same way as before with the sheaf $\mathcal{O}_V(*D)$, and so are morphisms between connections.
A fundamental initial observation is the following:

**Proposition V.5 ([HTT08], Lemma 5.3.1).**

The category of flat connections on $X$ is equivalent to the category of meromorphic flat connections with poles along $D$.

This means that contrary to the analytic case where Deligne had to define some conditions, the lifting of a connection on $X$ is unique (up to isomorphism) in the algebraic setup: it is based on the equality $\mathcal{O}_V(*D) = j_*\mathcal{O}_X$. Therefore, we only consider connection on $X$.

One of the main points here is to find a suitable definition of regularity for an integrable connection $(\mathcal{E}, \nabla)$ on $X$.
As the GAGA theorems work mainly on projective varieties, the first idea is to embed $X$ into a projective one before applying our tools and restricting to $X$. It can be done by the Hironaka's resolution of singularities ([Hir64]): there exists an open embedding $j : X \to \bar{X}$ into a projective $\bar{X}$ such that $Y := \bar{X} \setminus X$ is a normal crossing divisor. We thus define the following:

**Definition V.1.**

$(\mathcal{E}, \nabla)$ is said to be *regular* if $((j_*\mathcal{E})^{\mathrm{an}}, \nabla^{\mathrm{an}})$ is a regular meromorphic connection with poles along $Y^{\mathrm{an}}$, in the analytic sense.
We denote by $\mathrm{Conn}_{\mathrm{reg}}(X)$ the category of regular flat connections.

This definition depends *a priori* on the embedding $\bar{X}$ chosen, but one can show that it actually does not, see [[Bor+87], §7.1].
We can provide an equivalent formulation, not involving an embedding. It is this definition on which is based [HTT08]:

**Proposition V.6 ([Bor+87], Proposition 7.1.1).**

$(\mathcal{E}, \nabla)$ is regular if, and only if for any morphism $i_C : C \to X$ from an algebraic smooth curve $C$, the connection $i_C^*(\mathcal{E}, \nabla)$ is regular.

With some additional work, and the use of GAGA V.2 among other things, we obtain the following:

**Theorem V.7 ([Bor+87], Theorem 7.2.1 or [HTT08], Theorem 5.3.8).**

Let $X$ be a smooth algebraic variety.

The analytification functor induces an equivalence of categories:

$$\mathrm{Conn}_{\mathrm{reg}}(X) \simeq \mathrm{LocFree}^{\nabla}(X^{\mathrm{an}})$$

And directly, by the analytic case IV.23:

**Corollary V.8.**

Let $X$ be a smooth algebraic variety. Then there is an equivalence:

$$\mathrm{Conn}_{\mathrm{reg}}(X) \simeq \mathrm{LocSys}(X^{\mathrm{an}}).$$

In particular, we obtain a totally algebraic description of the fundamental group of $X^{\mathrm{an}}$.

# VI A higher generalization : the $\mathcal{D}$-module theory

This section basically follows [HTT08], trying to cover from chapter 1 to chapter 7.
We develop the notion of *analytic $\mathcal{D}$-modules*, meaning, $\mathcal{D}$-modules on a complex manifold, as opposite to *algebraic $\mathcal{D}$-modules*, on a (smooth) algebraic variety. The two are pretty similar, as explained in introduction of [[HTT08], Chapter 4]. We will explain how to switch to the algebraic setup, using the GAGA machinery.
The goal of this part is to be able to translate the tools seen so far in the algebraic setup of the $\mathcal{D}$-modules theory and state the corresponding Riemann-Hilbert correspondence.

## VI.1 General theory of $\mathcal{D}$-modules

Following the introduction of [HTT08], let us explain why $\mathcal{D}$-modules emerge from the theory of linear partial differential equations.

Let $U \subset \mathbb{C}^n$ be an open with coordinates $(x_1, \ldots, x_n)$ and denote by $D$ the set of differential operators with coefficients in $\mathscr{O} := \mathscr{O}_X(U)$:

$$D = \left\{ \sum_{i_1,\ldots,i_n}^{<\infty} a_{i_1,\ldots,i_n} \partial_1^{i_1} \ldots \partial_n^{i_n} \mid a_{i_1,\ldots,i_n} \in \mathscr{O} \right\}.$$

$\mathscr{O}$ is a left $D$-module, $D$ acting by derivation.
For $P \in D$, we consider the differential equation:

$$Pu = 0.$$

We associate to this equation the left $D$-modules $M = D/DP$, and one computes:

$$\mathrm{Hom}_D(M, \mathscr{O}) \simeq \{\varphi \in \mathrm{Hom}_D(D, \mathscr{O}) \mid \varphi(P) = 0\} \simeq \{f \in \mathscr{O} \mid Pf = 0\}.$$

We can replace in the left hand side $\mathscr{O}$ with another function space admitting a left-$D$-action.

This idea can be generalized to systems of linear partial differential equations, see [HTT08] or [Sch19] (both in the introduction) for details.
Therefore, an analytic problem is turned into an algebraic one. The theory of moving from one to the other is often refered as "algebraic analysis".

### VI.1.1 Definition and relation with connections

Let $X$ be a complex manifold of dimension $n$.
The sheaf of differential operators $D_X$ is the subring of $\mathcal{E}nd_{\mathbb{C}}(\mathscr{O}_X)$ generated by $\mathscr{O}_X$ (identified to a subring of $\mathcal{E}nd_{\mathbb{C}}(\mathscr{O}_X)$ via $f \mapsto (g \mapsto fg)$) and $\Theta_X$, the sheaf of vector field over $X$ (which acts locally by: if $\theta = \sum_i a_i \partial_i$ and $f \in \mathscr{O}(U)$, then $\theta f = \sum_i a_i \frac{\partial f}{\partial x_i}$).
With local coordinates $(x_1, \ldots, x_n)$ over $U$, we have

$$D_X|_U = \bigoplus_{\alpha \in \mathbb{N}^n} \mathscr{O}_U \partial^\alpha,$$

with $\partial^\alpha = \partial_1^{\alpha_1} \ldots \partial_n^{\alpha_n}$., see [[HTT08], I§1.1].

**Definition VI.1.**

We denote by  the category of $D_X$-modules.

From this "algebraic" construction, we can recover the analytic one described in the last two sections: the integrable connections.

**Definition VI.2.**

A $D_X$-module $M$ is said to be an *integrable connection* if $M$ is locally free ever $\mathscr{O}_X$ of finite rank.
We denote by $\mathrm{Conn}(X)$ the category of integrable connection.

**Remark VI.1.**

Let $M$ be an integrable connection (in the sense of the above definition).
The condition of the definition gives that $M$ is a vector bundle on $X$ (via the equivalence III.16).
The "connection" as a map (as in the previous sections) is recover via the action of $D_X$, by:

$$\nabla(\theta \otimes s) := \theta s$$

for $\theta \in \Theta_X$ and $s \in M$. It defines a map $\nabla : M \to \mathcal{E}nd_{\mathbb{C}}(\Theta_X, M) \simeq M \otimes_{\mathscr{O}_X} \Omega^1_X$, and one verifies that it is $\mathbb{C}$-linear and satisfies the Leibniz relation (see [HTT08], Lemma 1.2.1 for the details).

A important result garantying the "finiteness" number of equations of "the base" of our systems is the following.

**Proposition VI.1 ([HTT08], Theorem 4.1.2).**

$D_X$ is a coherent sheaf of rings.

**Remark VI.2.**

Here, coherent means: every finitely generated ideal is of finite presentation, *ie* the relation satisfied by its elements are finitely generated.
For the algebraic setup, this result directly comes from the fact that $\mathcal{O}_X$ is locally noetherian, and so is $D_X$.
It is not the case anymore for the analytic case. However, the stalks are indeed noetherian, see [[HTT08], Theorem 4.1.2].

The $D_X-modules$ in which we are interested for now on are also supposed to be coherent $D_X$-modules, and we denote this category as $\mathrm{Mod_c}(D_X)$.

### VI.1.2 Holonomicity

The notion of holonomy is mainly motivated by the constructibility theorem VI.4. The main idea is to recover the "finite dimension" of the space of solutions of $D_X$-modules.
For this goal, we consider the *characteristic variety*. We will not introduce it formally as it would require a background of *filtration theory*. See [[HTT08], §2.2] for the proper definition.

The idea is the following: for a differential operator $P$, one can construct its *principal symbol* $\sigma(P)$: it is a polynomial function on the cotangeant bundle $T^*X$, that keep the information of the "complexity" of the $D_X$-modules $M$ considered.
$M$ is "composed" of a system of such $P$, and one defines the *characteristic variety* $\mathrm{Ch}(M)$ as the set of all the principal symbols $\sigma(P)$.
It is known (see [[HTT08], Corollary 2.3.2]) that $\dim(\mathrm{Ch}(M)) \geq \dim(X)$ for a coherent $D_X$-module $M$. The case where the equality holds is our main concern:

**Definition VI.3.**

A coherent $D_X$-module $M$ is said to be *holonomic* if it satisfies the inequality:

$$\dim(\mathrm{Ch}(M)) \leq \dim(X).$$

We denote by $\mathrm{Mod}_{\mathrm{h}}(D_X)$ the full subcategory of $\mathrm{Mod}_{\mathrm{c}}(D_X)$ of holonomic $D_X$-modules.

While the coherence garanteed the finiteness of the number of equations, the holonomicity asks for a sufficient number of them such that it corresponds to the equations we encountered in the previous statements of the Riemann-Hilbert correpondence.
Therefore, a "coherent holonomic" $D_X$-modules has a "satisfying" number of equations.

### VI.1.3 Regularity

The last notion to be defined in order to get a complete analoguous to the correspondence stated before is the regularity.
The definition (and equivalent ones) are given in the case of algebraic and analytic $\mathcal{D}$-modules in [[HTT08], Chapter 6], and a link is made between both (in the sense of V.1).
In particular, a "curve testing criterion" still holds as V.6.

Again, we will not state the definitions, as they would need some additional preliminaries.
We denote by $\mathrm{Mod}_{\mathrm{rh}}(D_X)$ the full subcategory of $\mathrm{Mod}_{\mathrm{h}}(D_X)$ consisting of regular $D_X$-modules.

## VI.2 Solutions and de Rham functors

### VI.2.1 The necessity of derived categories

Based on what have been said in the introduction of VI.1, we would like to define the functor of solutions as:

$$\mathrm{Sol}_X M := \mathcal{H}om_{D_X}(M, \mathscr{O}_X),$$

for a $M$ an object of $\mathrm{Mod}(D_X)$.
The main problem to this definition is the non-right-exactness of the Hom functor. Therefore, some interesting functors on $D_X$-modules (such as pull-back, push-forward, etc.) do not commute with this definition of $\mathrm{Sol}_X$. Even though we do not develop the definition and the properties of these particular functors, they are useful in the proof of the Riemann-Hilbert correspondence that is to come.
To remedy this problem, we consider the derived categories of $D_X$-modules (which is an abelian category), and particularly its bounded subcategory, denoted by $\mathrm{D}^b(D_X)$.
In this category, the solutions functor becomes

$$\mathrm{Sol}_X M^\bullet := \mathrm{R}\mathcal{H}om_{D_X}(M^\bullet, \mathscr{O}_X),$$

and is exact.

Therefore, in this section, we translate all the categories and functors in terms of the derived formalism. In particular, we denote by $\mathrm{D}^b_{\mathrm{h}}(D_X)$ (resp. $\mathrm{D}^b_{\mathrm{rh}}(D_X)$, resp. $\mathrm{D}^b_{\mathrm{c}}(D_X)$) the subcategory of $D^b(D_X)$ whose cohomology groups are in $\mathrm{Mod}_{\mathrm{h}}(D_X)$ (resp. $\mathrm{Mod}_{\mathrm{rh}}(D_X)$, resp. $\mathrm{Mod}_{\mathrm{c}}(D_X)$). We assume the latters to be abelian categories.
The last theorem of this section restrict the Riemann-Hilbert correspondence on the derived categories to its subcategory of complexes concentrated at 0, and then recover a correspondence for $\mathrm{Mod}(D_X)$ itself.

### VI.2.2 The equivalence functors on a complex manifold

The solution $\mathcal{H}om_{D_X}(M, \mathscr{O}_X)$ of a $D_X$-module $M$ is a $\mathbb{C}_X$-module. In the bounded derived categories, $\mathrm{Sol}_X(M^\bullet) \in \mathrm{D}^b(\mathrm{Mod}(\mathbb{C}_X))$. The later category will simply be denoted as $\mathrm{D}^b(\mathbb{C}_X)$.
However, one can compute that the functor obtain is

$$\mathrm{Sol}_X : \mathrm{D}^b(D_X) \to \mathrm{D}^b(\mathbb{C}_X)^{\mathrm{op}},$$

with the opposite category in the co-domain. For the convenience, it will be intersting to introduce the *de Rham functor* as follows:

$$\mathrm{DR}_X : \mathrm{D}^b(D_X) \to \mathrm{D}^b(\mathbb{C}_X),$$
$$\mathrm{DR}_X(M^\bullet) = \Omega_X \otimes^{\mathrm{L}}_{D_X} M^\bullet,$$

where $\Omega_X := \bigwedge^{\dim(X)} \Omega^1_X$ is the *canonical sheaf* (see [HTT08], page 19) and $\Omega_X \otimes^{\mathrm{L}}_{D_X} -$ stands for the left derived functor associated to $\Omega_X \otimes_{D_X} -$.
Both are linked with the duality functor defined as:

$$\mathbb{D}_X : \mathrm{D}_{\bullet} tc^b(D_X) \to \mathrm{D}^b_{\mathrm{c}}(D_X),$$

$$\mathbb{D}_X M^\bullet = \mathrm{R}\mathcal{H}om_{D_X}(M^\bullet, D_X \otimes_{\mathscr{O}_X} \Omega_X^{\otimes -1}[\dim(X)]),$$

where $[\dim(X)]$ means the shifted-to-the-left-$\dim(X)$-times complex.
The rude definition does not really interest us. The following property however reveal the dual character of the $\mathrm{Sol}_X - \mathrm{DR}_X$ couple of functors:

**Proposition VI.2 ([HTT08], Proposition 4.2.1).**

Let $M^\bullet \in \mathrm{D}^b_{\mathrm{c}}(D_X)$.
$$\mathrm{DR}_X(M^\bullet) \simeq \mathrm{Sol}_X(\mathbb{D}_X M^\bullet)[\dim(X)].$$

The computation are usually easier with $\mathrm{DR}_X$. For example, considering a correct resolution of $\Omega$ and using the Poincaré's lemma, one can recover the classical Riemann-Hilbert correspondence IV.10 in this formalism:

**Proposition VI.3 ([HTT08], Theorem 4.2.4).**

Let $M$ be an integrable connection on $X$, and set $d = \dim(X)$.
We have an equivalence of categories:

$$\mathrm{H}^{-d}(\mathrm{DR}_X(-)) : \mathrm{Conn}(X) \xrightarrow{\sim} \mathrm{LocSys}(X).$$

### VI.2.3 The equivalence functors on a smooth algebraic variety

We use the GAGA paradigm to translate what we have been doing on an algebraic variety.
Let $X$ be a complex algebraic variety.
The first step is to establish a link between $D_X$-modules and $D_{X^{\mathrm{an}}}$-modules. From the morphism of ringed space $\iota : (X^{\mathrm{an}}, \mathscr{O}_{X^{\mathrm{an}}}) \to (X, \mathcal{O}_X)$ given by V.1, we have

$$D_{X^{\mathrm{an}}} \simeq \mathscr{O}_{X^{\mathrm{an}}} \otimes_{\iota^{-1}\mathcal{O}_X} \iota^{-1}D_X \simeq \iota^{-1}D_X \otimes_{\iota^{-1}\mathcal{O}_X} \mathscr{O}_{X^{\mathrm{an}}},$$

that enable us to define

$$(-)^{\mathrm{an}} : \mathrm{Mod}(D_X) \to \mathrm{Mod}(D_{X^{\mathrm{an}}}),$$
$$M^{\mathrm{an}} = D_{X^{\mathrm{an}}} \otimes_{\iota^{-1}D_X} \iota^{-1}M.$$

This functor extends to the derived categories:

$$(-)^{\mathrm{an}} : \mathrm{D}^b(D_X) \to \mathrm{D}^b(D_{X^{\mathrm{an}}}).$$

We define the solution and de Rham functors on $X$ as follows:

$$\mathrm{DR}_X : \mathrm{D}^b(D_X) \to \mathrm{D}^b(\mathbb{C}_X),$$

$$\mathrm{DR}_X(M^\bullet) = \mathrm{DR}_{X^{\mathrm{an}}}((M^\bullet)^{\mathrm{an}})$$

and

$$\mathrm{Sol}_X : \mathrm{D}^b(D_X) \to \mathrm{D}^b(\mathbb{C}_X)^{\mathrm{op}},$$
$$\mathrm{DR}_X(M^\bullet) = \mathrm{Sol}_{X^{\mathrm{an}}}((M^\bullet)^{\mathrm{an}}).$$

The verification of the compatibility of this functor with de numerous functor on $\mathcal{D}$-modules is done in [[HTT08], §4.7]. In particular, the duality VI.3 still hold.

## VI.3 The Riemann-Hilbert correspondence for $\mathcal{D}$-modules

### VI.3.1 Constructibility, perversity

We describe the essential image of the de Rham functor in this subsubsection.
We want to generalize the notion of local system. It is done in the following way:

**Definition VI.4.**

Let $X$ be an analytic space (resp. an algebraic variety), $X = \bigsqcup_{\alpha \in A} X_\alpha$ a locally finite partition of $X$ by locally closed (*ie* $X_\alpha$ is the intersection of an open and a closed set of $X$) analytic subset (resp. algebraic subvariety) of $X$.
This partition is called a *stratification* if $X_\alpha$ is smooth, (*ie* is a complex manifold for the analytic case), and if for each $\alpha \in A$, there exists $B_\alpha \subset A$ such that:

$$\bar{X}_\alpha = \bigsqcup_{\beta \in B_\alpha} X_\beta.$$

Each $X_\alpha$ is called a *stratum* of the stratification.

**Definition VI.5.**

Let $X$ be an analytic space.
A $\mathbb{C}_X$-module $F$ is called a *constructible sheaf of* $X$ if there exists a stratification $X = \bigsqcup_{\alpha\in A} X_\alpha$ such that for every $\alpha \in A$, $F|_{X_\alpha}$ is a local system.

**Example VI.3.**

As the notion can be abrupt, we give a simple example.
Let us take our favorite differential equation $(x\frac{d}{dx} - \lambda)u = 0$, for a complex number $\lambda$, on $X = \mathbb{C}$.
Therefore, the sheaf of holomorphic solutions $F \subset \mathscr{O}_{\mathbb{C}}$ is constructible with respect to the stratification $\{0\} \sqcup \mathbb{C}^*$: on $\mathbb{C}^*$, the solutions are of the form $x \mapsto Ax^\lambda$ for $A \in \mathbb{C}$ and at 0, and the stalk is computed to be $\mathbb{C}$ if $\lambda \in \mathbb{N}$ and 0 else (depending on the multivaluedness of the solutions: $x\ mapstox^\lambda = e^{\log(x)\lambda}$ is well defined on an open neighbourhood of 0 if, and only if $\lambda \in \mathbb{N}$).

For an algebraic variety, we reduce it to the analytic case:

**Definition VI.6.**

Let $X$ be an algebraic variety.
A $\mathbb{C}_{X^{\mathrm{an}}}$-module is said to be an *algebraically constructible sheaf* if there exists a stratification $X = \bigsqcup_{\alpha\in A} X_\alpha$ such that for every $\alpha \in A$, $F|_{X_\alpha^{\mathrm{an}}}$ is a locally constant sheaf.

For an analytic space (resp. an algebraic variety) $X$, we denote by $\mathrm{D}^b_{\mathrm{c}}(X)$ (resp. $\mathrm{D}^b_{\mathrm{c}}(X)$) the full subcategory of $\mathrm{D}^b(\mathbb{C}_X)$ (resp. $\mathrm{D}^b(\mathbb{C}_{X^{\mathrm{an}}})$) whose cohomology groups are constructible sheaves (resp. algebraically constructible sheaves).
That said, we can state the theorem that link holonomicity with constructibility:

**Theorem VI.4 (Kashiwara's constructibility, [HTT08], Theorem 4.6.3 and Theorem 4.7.7).**

Let $X$ be a complex manifold or an algebraic variety, $M^\bullet \in \mathrm{D}^b_{\mathrm{h}}(D_X)$.
Then $\mathrm{Sol}_X(M^\bullet)$ and $\mathrm{DR}_X(M^\bullet)$ are in $\mathrm{D}^b_{\mathrm{c}}(X)$.

The image of the de Rham functor on $\mathrm{Mod}_{\mathrm{h}}(D_X)$, ie complexes concentrated at 0 of $\mathrm{D}^b_{\mathrm{h}}(D_X)$ has a sharper description:

**Proposition VI.5 ([HTT08], Theorem 4.6.6).**

Let $X$ be a complex manifold and let $M \in \mathrm{Mod}_{\mathrm{h}}(D_X)$.
Then for every $j \in \mathbb{Z}$,

$$\dim(\mathrm{Supp}(H^j(\mathrm{DR}_X(M))) \leq -j \text{ and } \dim(\mathrm{Supp}(H^j(\mathbf{D}_X\mathrm{DR}_X(M))) \leq -j,$$

for another duality functor $\mathbf{D}_X$ that we will not define, see [[HTT08], Theorem 4.5.2] for one.

In fact, these conditions suffices, in the sense of VI.8.

**Definition VI.7.**

An object $F^\bullet \in \mathrm{D}^b_{\mathrm{c}}(X)$ is called a *perverse sheaves* if it satisfies the above conditions, namely for every $j \in \mathbb{Z}$,

$$\dim(\mathrm{Supp}(H^j)) \leq -j \text{ and } \dim(\mathrm{Supp}(H^j \mathbf{D}_X F^\bullet)) \leq -j.$$

We denote by $\mathrm{Perv}(\mathbb{C}_X)$ the full subcategory of $\mathrm{D}^b_{\mathrm{c}}(X)$ consisting of perverse sheaves.

### VI.3.2 Statements of the equivalence

The first theorem "Riemann-Hilbert" in the language of $\mathcal{D}$-modules was proved by Kashiwara in 1980 (he constructed an explicit inverse functor using the theory of distributions), and some years lated by Mebkhout (with more algebraic methods).

**Theorem VI.6 ([HTT08], Theorem 7.2.1).**

Let $X$ be a complex manifold.
The de Rham functor induces an equivalence of categories:

$$\mathrm{D}^b_{\mathrm{rh}}(D_X) \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{c}}(X).$$

After the development of the algebraic $\mathcal{D}$-modules, an analoguous could be stated for algebraic varities:

**Theorem VI.7 ([HTT08], Theorem 7.2.2).**

Let $X$ be a complex algebraic variety.
The de Rham functor induces an equivalence of categories:

$$\mathrm{D}^b_{\mathrm{rh}}(D_X) \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{c}}(X).$$

Finally, the restriction of this functor to complexes concentrated at 0 has the following image, proven by Kashiwara in his PhD in 1975 (before that the theory of perverse sheaves have been really developed in 1982, see [BBD82]).

**Theorem VI.8 ([HTT08], Theorem 7.2.5).**

Let $X$ be a smooth algebraic variety.
The de Rham functor induces an equivalence of categories:

$$\mathrm{Mod}_{\mathrm{rh}}(D_X) \xrightarrow{\sim} \mathrm{Perv}(\mathbb{C}_X).$$

This is considered to one of the most complete solution to Hilbert 21st problem.

# VII Periods

## VII.1 Prerequisites of periods theory

We will not be able to provide an introduction as appetizing as the theory of periods would call for. A document of Javier Fresán [Fre22] (in French) gives a historical approach (starting with Newton), and some modern developments of the theory.
Here, we only give the necessary knowledge to read the rest of the report.
We use [Fre22] and [HM17].

### VII.1.1 A first approach

Period numbers form a class of numbers, bigger than $\bar{\mathbb{Q}}$, that is supposed to contain *every reasonable number* that one could encounter. The designation originally comes from the period of an oval, or the period of a pendulum.
For the first definition, we need the following.

> **Definition VII.1.**
>
> Let a set $S \subset \mathbb{R}^n$.
> $S$ is said to be $\mathbb{Q}$-*semi-algebraic* if it can be expressed with finitely many intersection, union or complement of sets of the form $\{(x_1, \dots, x_n), P(x_1, \dots, x_n) \geq 0\}$ with $P \in \mathbb{Q}[X_1, \dots, X_n]$.

> **Definition VII.2.**
>
> A complex number $Z$ is said to be a *period* if its real part and its imaginary part can be expressed as absolutely convergent integrals with respect to the Lebesgue measure $\mu$
>
> $$\int_S f(x_1, \dots, x_n) d\mu(x_1, \dots, x_n),$$
>
> with $S$ a $\mathbb{Q}$-semi-algebraic set and $f \in \mathbb{Q}(x_1, \dots, x_n)$.
> We denote by $\mathcal{P} \subset \mathbb{C}$ the set of all periods.

$\mathcal{P}$ is clearly countable, so almost-every complex number is not a period.
Algebraically :

> **Theorem VII.1 ([Fre22], Lemma 2.2).**
>
> $\mathcal{P}$ is a subring of $\mathbb{C}$.

Some examples are elements in $\bar{\mathbb{Q}}$, $\pi$, (multi-)Zeta values on the integer $\zeta(n)$, $\log(q)$ for positive $q \in \mathbb{Q}$ ... Actually, finding a number that is *not to be* a period is difficult.

Period numbers can have different integral representations, for example

$$\pi = \int_{x^2+y^2 \leq 1} dxdy = \int_{\mathbb{R}} \frac{dx}{x^2+1}.$$

We cannot proceed without stating of the most important conjecture in the field (not very precisely though, see [[Fre22], §3.1] for a more precise statement) :

**Conjecture VII.2 (Kontsevich-Zagier, 2001, [Fre22], Conjecture 3.1).**

Every linear relation between periods comes from these operations on (absolutely convergent) integrals :

1) Linearity,
2) change of variables,
3) stokes formula.

### VII.1.2 The cohomological perspective

We will not really motivate the introduction of cohomological tools used for the study of periods (partly because I do not know it very well myself). Some good references for it seem to be [[Fre22], §3.3], [And16] (both in French).

Let $X$ be an algebraic variety over a field $k \subset \mathbb{C}$ and denote by $X_{\mathbb{C}}$ the complexification of $X$. We use the isomorphisms between the de Rham, the singular and the sheaf cohomology given in III.5 for make appears the periods.

**Definition VII.3 ([HM17], Definition 5.3.1).**

We define the period isomorphism

$$\mathrm{comp} : \mathrm{H}^{\bullet}_{\mathrm{dR}}(X) \otimes_k \mathbb{C} \to \mathrm{H}^{\bullet}_{\mathrm{sing}}(X, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

as the composition of :

- $\mathrm{H}^{\bullet}_{\mathrm{dR}} \otimes_k \mathbb{C} \to \mathrm{H}^{\bullet}_{\mathrm{dR}}(X_{\mathbb{C}})$ (III.26)
- $\mathrm{H}^{\bullet}_{\mathrm{dR}}(X_{\mathbb{C}}) \to \mathrm{H}^{\bullet}_{\mathrm{dR}}(X^{\mathrm{an}})$ (VII.6)
- $\mathrm{H}^{\bullet}_{\mathrm{dR}}(X^{\mathrm{an}}) \to \mathrm{H}^{\bullet}_{\mathrm{sing}}(X^{\mathrm{an}}, \mathbb{C})$ (III.30)
- $\mathrm{H}^{\bullet}_{\mathrm{sing}}(X^{\mathrm{an}}, \mathbb{C}) \to \mathrm{H}^{\bullet}_{\mathrm{sing}}(X^{\mathrm{an}}, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$ (III.29)

given by

$$\mathrm{comp}([\omega]) : [\sigma] \mapsto \int_{\sigma} \omega,$$

together with the *period pairing* :

$$\begin{array}{rccl} \mathrm{per} : & \mathrm{H}^i_{\mathrm{dR}}(X) \times \mathrm{H}_i^{\mathrm{sing}}(X^{\mathrm{an}}, \mathbb{Q}) & \longrightarrow & \mathbb{C} \\ & ([\omega], [\sigma]) & \longmapsto & \int_{\sigma} \omega \end{array}.$$

**Remark VII.1.**

This isomorphism gives a $\mathbb{Q}$-*structure* (meaning, a structure of $\mathbb{Q}$-vector space) on $\mathrm{H}^i_{\mathrm{dR}}(X) \otimes_k \mathbb{C}$ as the image $\mathrm{comp}^{-1}(\mathrm{H}^i_{\mathrm{sing}}(X, \mathbb{Q}) \otimes 1)$, that forms with the $k$-structure and some filtrations on the complexification, called *Hodge filtration*, a *Hodge structure* on $\mathrm{H}^{\bullet}_{\mathrm{dR}}(X)$.

This is the geometric translation of the first definition of periods. Advantages of this formulation is discussed at the end of this report.

## VII.2 The Gauss-Manin connection

### VII.2.1 Coming from topology

We essentially use [[Voi02], Chapter 9].

Let $\mathscr{X}$ and $S$ be complex manifolds.

**Definition VII.4.**

Let $\pi : \mathscr{X} \to S$ be a holomorphic map.
We say that $\mathscr{X} \xrightarrow{\pi} S$ is a *family of complex manifold* if $\phi$ is a proper submersion.

We recall that proper means that the inverse image of a compact set is compact.
We will denote by $\mathscr{X}_s$ the fiber $\pi^{-1}(s)$ above $s \in S$.
The following result gives us the differentiable local behavior of such a family.

**Theorem VII.3 (Ehresmann, [Voi02], Proposition 9.3).**

Let $\mathscr{X} \xrightarrow{\pi} S$ be a family of complex manifolds, and assume that $S$ is contractible, with a base point $O$.
Then, there exists a diffeomorphism $T : \mathscr{X} \xrightarrow{\sim} \mathscr{X}_O \times S$ such that $\mathrm{pr}_2 \circ T = \pi$.

**Remark VII.2.**

This theorem only gives information on the differentiable structures. Indeed, there exists counter-examples for the analytic structure, see [[Poo25], Warning 4.9]. It can however be improved in this case, see [[Voi02], Proposition 9.5].

Let $\mathscr{X} \xrightarrow{\pi} S$ be a family of complex manifold. By the so-called *proper base change theorem*, we obtain:

**Proposition VII.4 ([Dim04], Proposition 2.3.26).**

For every $s \in S$ and every $i \geq 0$, we have

$$(\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}})_s \simeq \mathrm{H}^i(\mathscr{X}_s, \mathbb{C}).$$

$S$ is a complex manifold, so it is locally contractible: for every $s \in S$, there exists an open neighbourhood $U$ of $s$ which is locally contractible. By Ehresmann's theorem, there exists an isomorphism $T : \pi^{-1}(U) \to \mathscr{X}_s \times U$, such that $\pi = \mathrm{pr}_2 \circ T$. Thus, we have

$$(\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}})|_U \simeq \mathrm{H}^i(\mathscr{X}_s \times U, \mathbb{C}) \simeq \mathrm{H}^i(\mathscr{X}_s, \mathbb{C}) \simeq (\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}})_s,$$

as $U$ is contractible, and the sheaf $\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}}$ is locally constant.
Therefore, $\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}}$ is a local system. By applying the Riemann-Hilbert correspondence, we get:

**Definition VII.5.**

The natural connection $1 \otimes d$ on $\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}} \otimes_{\mathbb{C}} \mathscr{O}_S$ is called the *Gauss-Manin connection associated to the family* $\mathscr{X} \xrightarrow{\pi} S$ and is denoted by $\nabla_{\mathrm{GM}}$.

### VII.2.2 Coming from algebraic constructions

In this subsubsection, $\mathscr{X} \xrightarrow{\pi} S$ is a family of complex varities, meaning that $\pi$ is smooth proper and $S$ is assumed to be smooth over $\mathbb{C}$.

#### VII.2.2.a. With the algebraic de Rham cohomology

This paragraph is an attempt to unpack ideas from [[Blo16], §1.3].

Consider the the sheaf relative 1-forms

$$\Omega^1_{\mathscr{X}/S} := \Omega^1_{\mathscr{X}}/\pi^*\Omega^1_S$$

and the associated $j$-forms

$$\Omega^j_{\mathscr{X}/S} := \bigwedge^j \Omega^1_{\mathscr{X}/S}.$$

They correspond to the $j$-forms over $\mathscr{X}$ "modulo the ones coming from $S$". These $j$-forms are equipped with natural maps $p_j : \Omega^j_{\mathscr{X}} \to \Omega^j_{\mathscr{X}/S}$.
From there, we can construct the *relative de Rham complex*

$$\Omega^\bullet_{\mathscr{X}/S} : 0 \to \mathcal{O}_X \xrightarrow{d} \Omega^1_{\mathscr{X}/S} \xrightarrow{d} \dots \xrightarrow{d} \Omega^n_{\mathscr{X}/S},$$

where the derivations are given by $d^j_{\mathscr{X}/S} = p_{j+1} \circ d^j_{\mathscr{X}}$.
The analytification of this complex gives a resolution of the sheaf $\pi^{-1}\mathscr{O}_{S^{\mathrm{an}}}$ (we take the same notation $\pi$ for $\pi^{\mathrm{an}}$). Indeed, we have the following:

$$\mathrm{Ker}(p_1 \circ d^0_{\mathscr{X}^{\mathrm{an}}}) = (d^0_{\mathscr{X}^{\mathrm{an}}})^{-1}(\mathrm{Ker}(p_1)) = (d^0_{\mathscr{X}^{\mathrm{an}}})^{-1}(\pi^*\Omega^1_{S^{\mathrm{an}}}) = \pi^{-1}(d^0_{\mathscr{X}^{\mathrm{an}}})^{-1}\Omega^1_{S^{\mathrm{an}}} = \pi^{-1}\mathscr{O}_{S^{\mathrm{an}}}.$$

Therefore, we have the equality

$$\mathrm{R}^\bullet\pi_*(\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}) \simeq \mathrm{R}^\bullet\pi_*(\pi^{-1}\mathscr{O}_{S^{\mathrm{an}}}).$$

By writing $\pi^{-1}\mathscr{O}_{S^{\mathrm{an}}} = \pi^{-1}\mathscr{O}_{S^{\mathrm{an}}} \otimes_{\pi^{-1}\mathbb{C}_S^{\mathrm{an}}} \mathbb{C}_{\mathscr{X}^{\mathrm{an}}}$ and using the projection formula, we get:

$$\mathrm{R}^\bullet\pi_*(\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}) \simeq \mathrm{R}^\bullet\pi_*(\mathbb{C}_{\mathscr{X}^{\mathrm{an}}}) \otimes_{\mathbb{C}_{S^{\mathrm{an}}}} \mathscr{O}_{S^{\mathrm{an}}}.$$

The left-hand side correpond to the *relative de Rham cohomology*, and is equipped with an integrable connection given on the right-hand side by $1 \otimes d$.

**Proposition VII.5.**

The above connection is the *Gauss-Manin connection* associated to $\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}$.

The connection coincides (or at least, there are isomorphic) with the first definition because of the Riemann-Hilbert correpondence and the fact that both associated local systems have the same fibers. This reformulation is rather useful because of a result by Grothendieck:

**Theorem VII.6 ([Gro66], Theorem 1).**

The relative algebraic and analytic de Rham cohomology groups are isomorphic. In formulas:

$$(\mathrm{R}^\bullet\pi_*\Omega^\bullet_{\mathscr{X}/S})^{\mathrm{an}} \simeq \mathrm{R}^\bullet\pi^{\mathrm{an}}_*\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}.$$

To descend the Gauss-Manin connection to the algebraic world, we still need to check that it is regular (and use V.7). This is a theorem of Griffiths:

**Theorem VII.7 (Griffiths, [Gri70], Theorem 4.3).**

The Gauss-Manin connection is regular.

Therefore, we have:

**Proposition VII.8.**

The relative algebraic de Rham cohomology is equipped with an algebraic Gauss-Manin connection.

### VII.2.2.b. Using filtrations

The construction of this paragraph is taken from [KO68], and some retrospective clarifications in [[Hul25], §2.3] and [[Blo16], §1.3].
The advantage is that we can get rid of the proper hypothesis (Katz and Oda do not assume it in the original construction).
Start with the exact sequence coming from the definition:

$$0 \to \pi^*\Omega^1_S \to \Omega^1_{\mathscr{X}} \to \Omega^1_{\mathscr{X}/S} \to 0.$$

It induces a filtration on $\Omega^\bullet_X$:

$$F^p\Omega^n_X := \operatorname{Im}\left(\pi^*\Omega^p_S \otimes_{\mathcal{O}_{\mathscr{X}}} \Omega^{n-p}_X \xrightarrow{\eta\otimes\alpha\mapsto\pi*\eta\wedge\alpha} \Omega^n_X\right).$$

By computing the terms of the exact short sequence:

$$0 \to F^1/F^2 \to F^0/F^2 \to F^0/F^1 \to 0,$$

we obtain

$$0 \to \Omega^\bullet_{\mathscr{X}/S}[-1] \otimes_{\mathcal{O}_{\mathscr{X}}} \pi^*\Omega^1_S \to \Omega^\bullet_{\mathscr{X}}/F^2 \to \Omega^\bullet_{\mathscr{X}/S} \to 0.$$

This is a distinguished triangle in the derived category. As we did not develop this vocabulary, we skip the computation, but we get at the end an application:

$$\nabla : \mathrm{R}^i\pi_*\Omega^\bullet_{\mathscr{X}/S} \to \mathrm{R}^i\pi_*\Omega^\bullet_{\mathscr{X}/S} \otimes_{\mathcal{O}_S} \Omega^1_S.$$

This corresponds again to the Gauss-Manin connection.

## VII.3 The variational theory of periods and the Picard-Fuchs equations

The situation is now the following : consider $k \subset \mathbb{C}$ a field, and a family $\mathscr{X} \xrightarrow{\pi} S$ of algebraic varieties over $k$.
We have at our disposal the complexified $\mathscr{X}_{\mathbb{C}} \xrightarrow{\pi_{\mathbb{C}}} S_{\mathbb{C}}$ and the analytified $\mathscr{X}^{\mathrm{an}} \xrightarrow{\pi^{\mathrm{an}}} S^{\mathrm{an}}$ over $\mathbb{C}$.
We study the variation of the period numbers on the fibers $\mathscr{X}_s$ along $S$, by the use the previously constructed Gauss-Manin connection.
The references are again fragments of [Blo16].

Consider a family $(\sigma_s)_{s\in S}$ of Betti classes and $(\omega_s)_{s\in S}$ of relative de Rham classes.
We obtain a function
$$F(s) = \int_{\sigma_s} \omega_s.$$
We want to understand the behavior of the variation of $F$.
Its exterior derivative is related to the Gauss-Manin in the following way :

**Proposition VII.9.**

$$d_S F(s) = \int_{\sigma_s} \nabla_{\mathrm{GM}}(\omega_s)$$

Proof

The comparison isomorphism in VII.2.2.a gives

$$\mathrm{R}^\bullet \pi_*(\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}) \simeq \mathrm{R}^\bullet \pi_*(\mathbb{C}_{\mathscr{X}^{\mathrm{an}}}) \otimes_{\mathbb{C}_{S^{\mathrm{an}}}} \mathscr{O}_{S^{\mathrm{an}}}.$$

In a local basis, this isomorphism has $(j,k)$ matrix coefficients equal to $\int_{\sigma_{j,s}} \omega_{k,s} =: \langle \sigma_{j,s}, \omega_{k,s}\rangle$ and the derivation acts by Leibniz as $d_S\langle \sigma_{j,s}, \omega_{k,s}\rangle = \langle \sigma_{j,s}, \nabla_{\mathrm{GM}}(\omega_{k,s})\rangle$ because $\sigma_{j,s}$ is a flat section.

□

For a first explicit approach, let us assume that $S$ is a curve. Therefore, noticing that the $\mathrm{H}^i_{\mathrm{dR}}(\mathscr{X}_s)$ are finite dimensional, say $n$, over $\mathbb{C}_s$, so there exists a linear relation between $(\omega_s, \ldots, \nabla^n_{\mathrm{GM}}(\omega_s))$, say :
$$\sum_{j=0}^{n} a_j(s) \nabla^j_{\mathrm{GM}}(\omega_s) = 0.$$
In the other side of the above equality, we get :
$$\sum_{j=0}^{n} a_j F^{(j)}(s) = \sum_{j=0}^{n} a_j \int_{\sigma_s} \nabla^j_{\mathrm{GM}}(\omega_s) = 0.$$
We then obtain a linear differential equation for $F$.

When $S$ is not a curve, we use the same idea as III.23, to obtain a system of partial differential equations.
Now that we have developed the formalism of $\mathcal{D}$-modules, we can write it in a nicer way.

**Remark VII.3.**

The $\mathcal{D}$-modules formalism allows us to compute the Gauss-Manin, and a theorem of commutation of the (informal) form $\mathrm{R}f_* \circ \mathrm{DR}_X \simeq \mathrm{DR}_S \circ \mathrm{R}f_+$, applied to the $D_X$-module $\mathscr{O}_X$, recover the comparison isomorphism, and provide a new point of view on periods.

# VIII Conclusion and follow-ups

The short amount of time of the internship did not allow the report to be *totally* self-contained, in the sense that the proofs given are mainly present in the part IV.2 and IV.3.1: there were the parts that I was originally told to detail.
I hope that the references I gave are sufficient.

Let us now outline some possible directions for future research projects.
<u>Methodological refinements</u>:

- First, the vocabulary for the tools already used, that could translate into modern terms. Algebraic varieties could have been replaced with *schemes* (the fundamental work of Deligne [Del70] is written in these formalism), some (co)homological tools (*Mayer-Vitoris sequence*, *spectal sequence)* to get a better grip on derived categories and cohomology theories.
- The (algebraic) $\mathcal{D}$-modules theory is way more vast and rich than the one that was presented, and for which I had the chance to get a glimpse of it thanks to the seminar.
- Various methods for the proof of the Riemann-Hilbert on $\mathcal{D}$-modules VI.3.2 have been developed (analytic by Kashiwara [Kas84], more algebraic by Mebkhout [Meb05] etc.), and understanding of them could help the grasp over the theorem.
  Moreover, this equivalence has been proven more recently to hold even for irregular systems using *ind-sheaves*, see for instance [KD24] for a presentation of this work.

<u>Broader arithmetic horizons</u>

- Related to the previous one, irregular $\mathcal{D}$-modules give rise to different exceptional complex number, as exponential periods (roughly speaking, of the form $\int_\sigma e^{-f}\omega$, where $\sigma$, $f$ and $\omega$ are of algebraic nature). The latter is also natural to be considered in a context of $o$-minimal structure, coming from a geometrico-logic framework of *tame geometry.* The articles [FJ17] and [CHH25] seems to be references.
- More globally, the theory of periods and their relation to *motives* under the general theory of a Galois theory for periods is a really active domain of research. It gives rise to a new formulation of the Kontsevich-Zagier conjecture, made by Grothendieck, more geometric (see for example [Fre22], Conjecture 10.1]).
- A general discussion on the differential equations of the form appearing in VII.3 (said to be *coming from a geometric origin*) are subjects to numerous theories and conjectures (Bombieri–Dwork, Grothendieck–Katz $p$-curvature ...), discussed in [And89]. This can be used in the theory of transcendence as I first became aware of these theories during an Arbeitsgemeinschaft on a recent paper by Calegari, Dimitrov and Tang [CDT24].
- A conterpart of the study of the variation of periods is the *Hodge theory*, examining the variation of the Hodge structure that gives information on the geometry of the varieties (see [Voi02]).
- Finally, all this framework can be translated in a more "arithmetic" world: the $p$-adic one - $p$-adic periods, $p$-adic Riemann-Hilbert correspondence, $p$-adic differential equations, $p$-adic Hodge theory, arithmetic $\mathcal{D}$-modules. As an amateur of $p$-adic numbers, these theories particularly resonate with me.
- Learning a bit of German would have been nice, maybe for a future, longer internship !

In personal point of view, this internship has confirmed my will to work in fundamental mathematics. This first step into the world of arithmetic geometry and the the number of possible paths ahead amazed me.
Fortunatly, I have been accepted (during the internship) to the Master's program in Lyon in Arithmetic

geometry, and some of my concern listed above will be adress. Some future research project will eventually treat in details some of them, in the internship at the end of the M2, and above all, for a potential PhD.

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# List of notations

## Section III :



## Section IV :



## Section V :



## Section VI :



## Section VII :